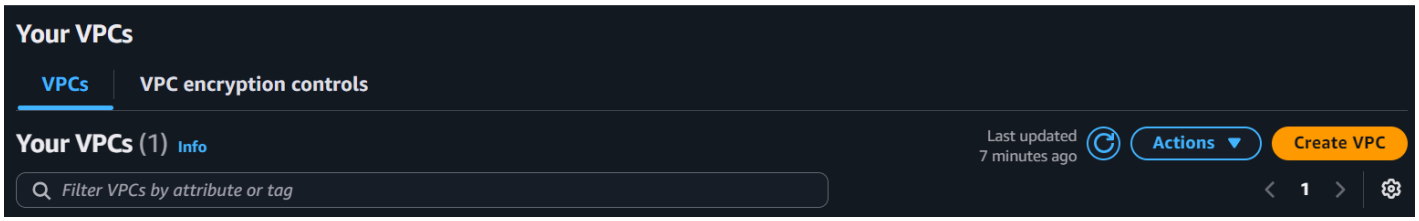
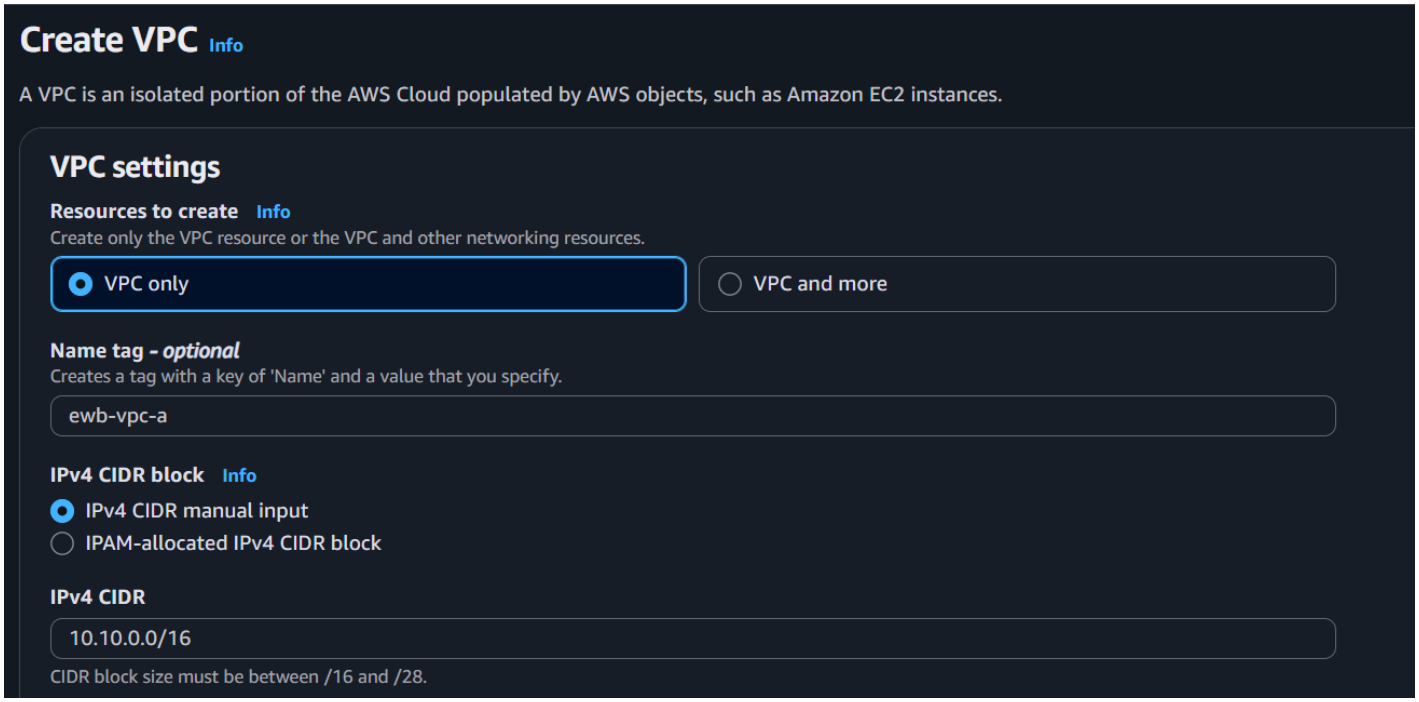
EWB
Courses

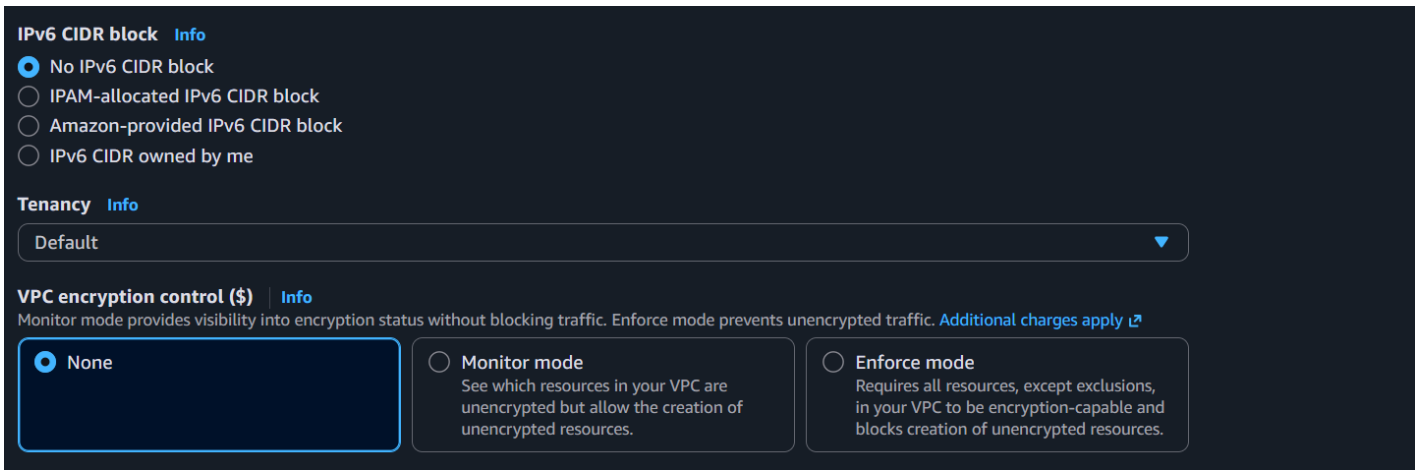
(Project)

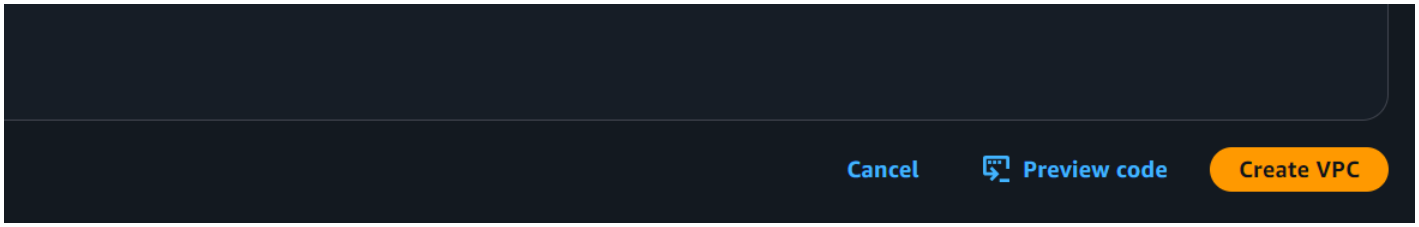
AWS VPC Peering

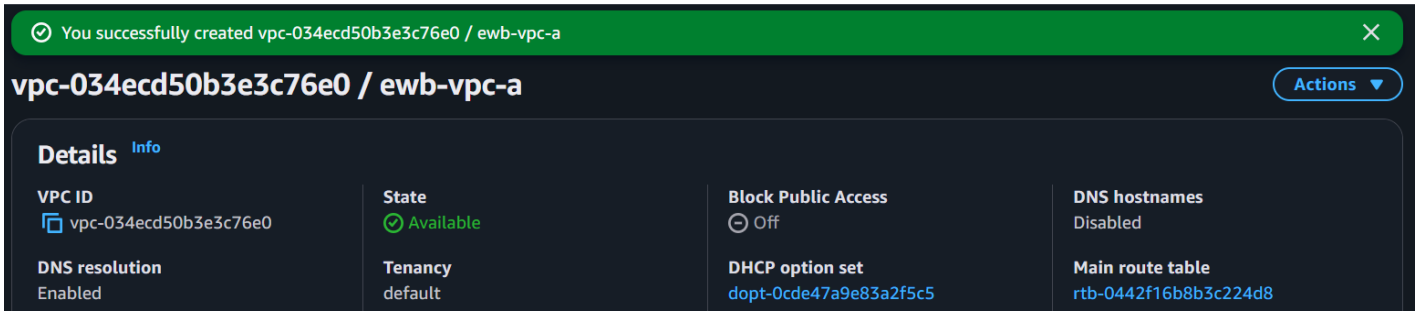
Step 1:- Create the VPC.



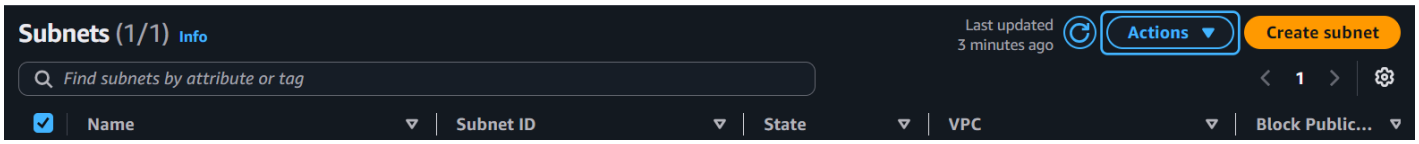


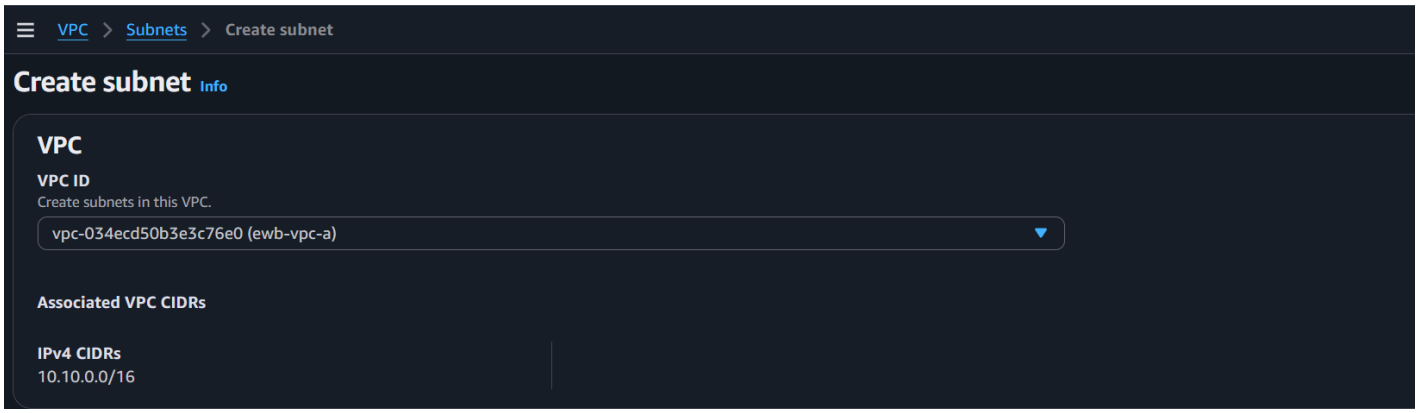


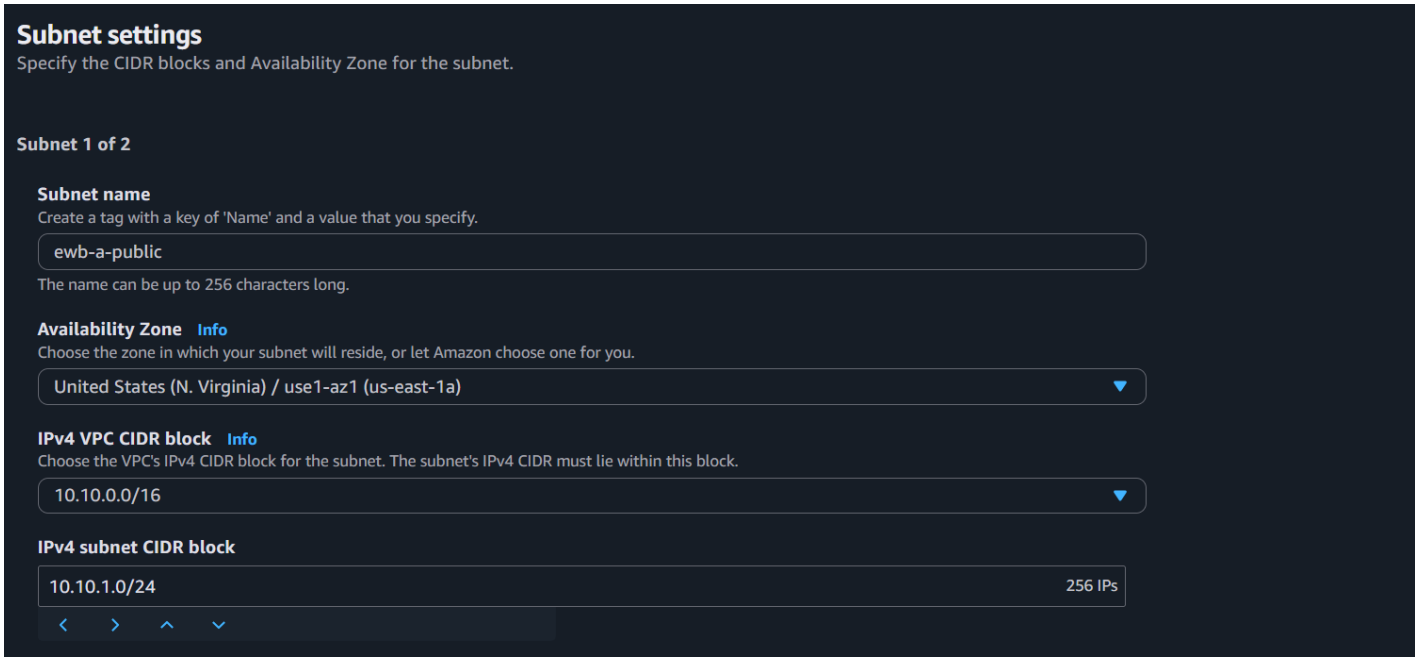




Step 2:- Create the subnets.

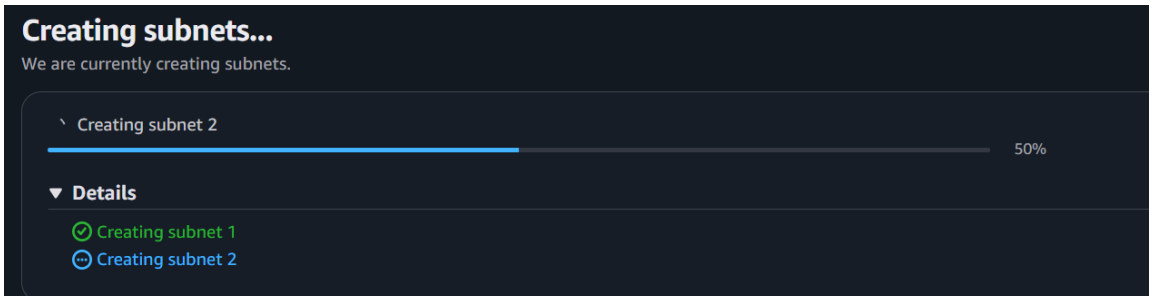


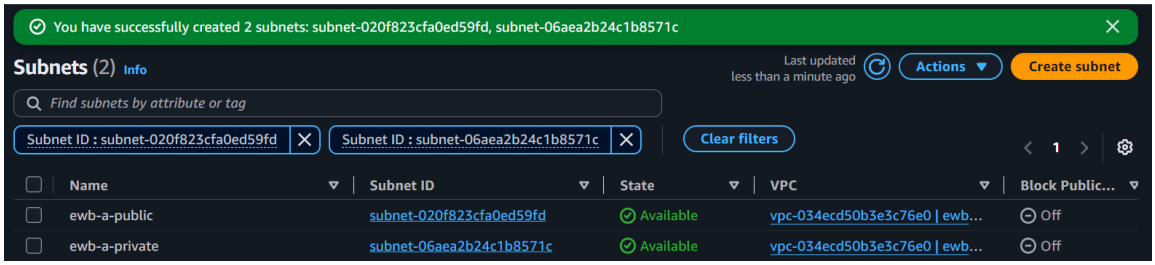




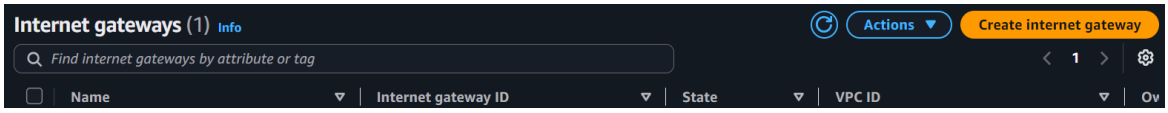


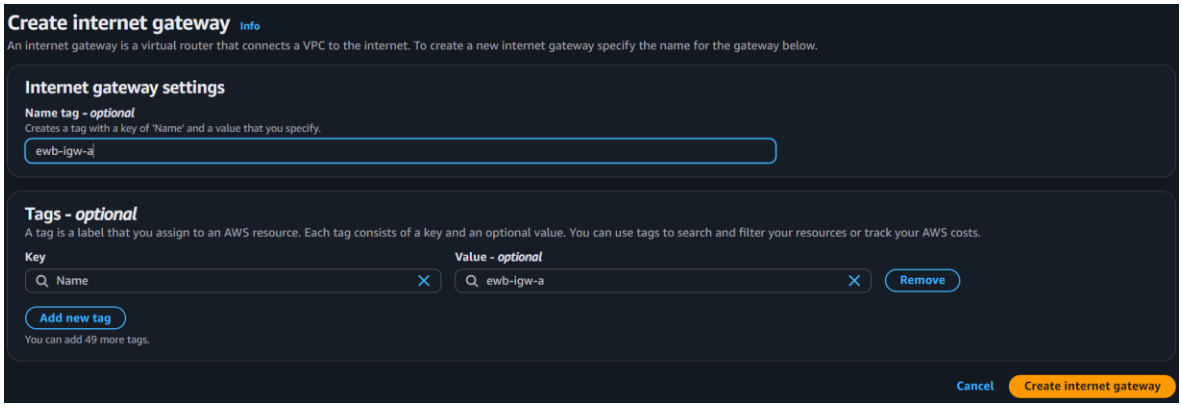


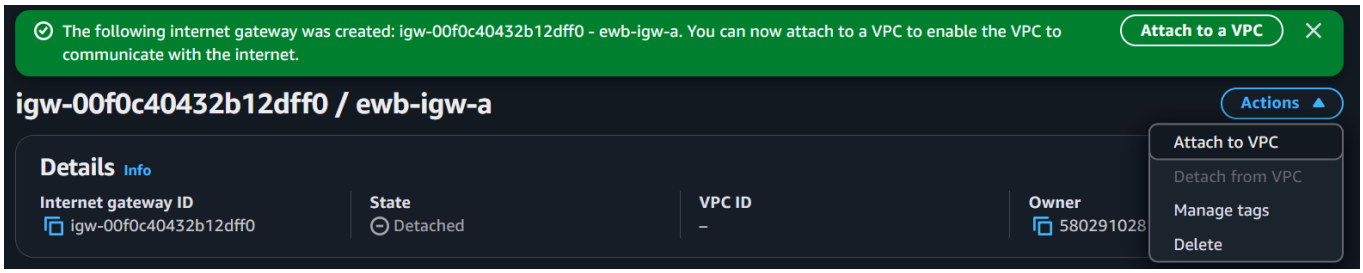


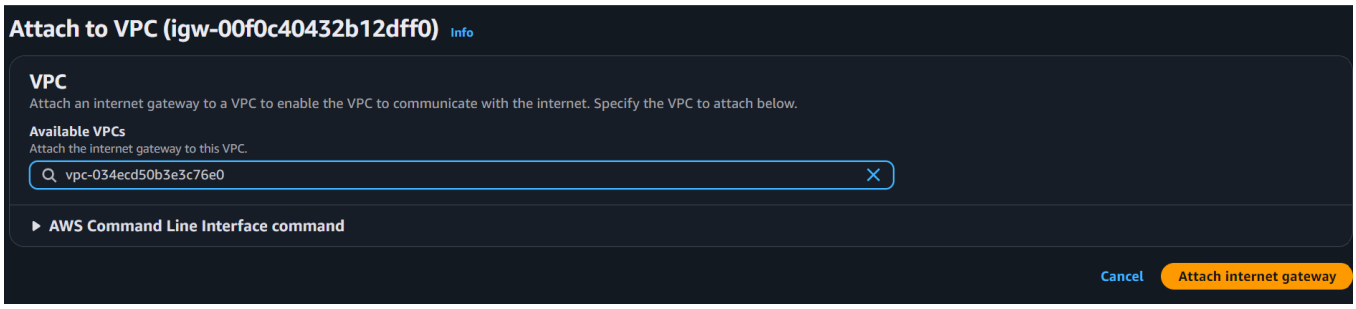


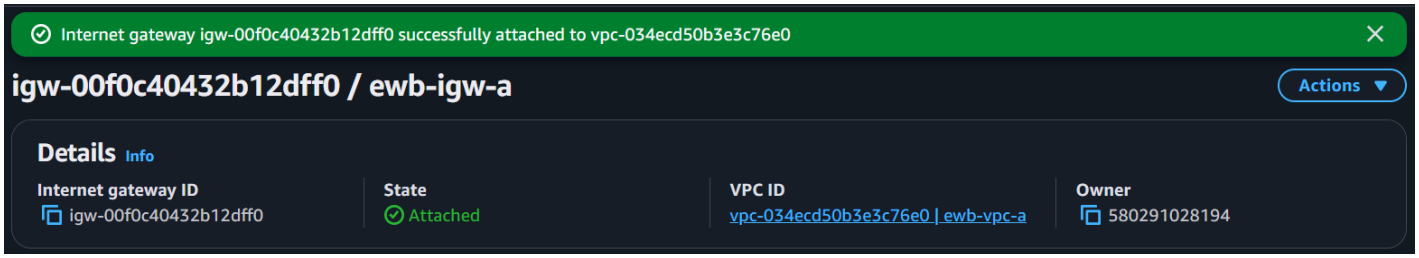
Step 3:- Create an Internet Gateway.



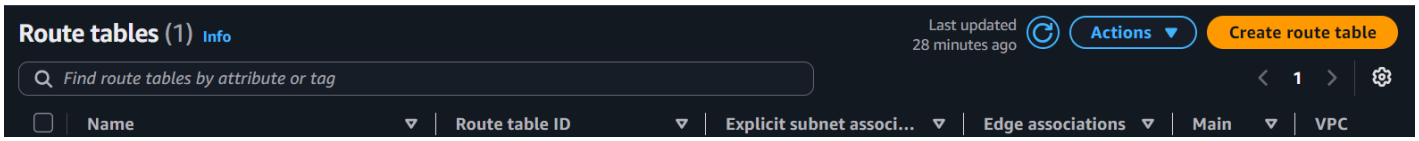


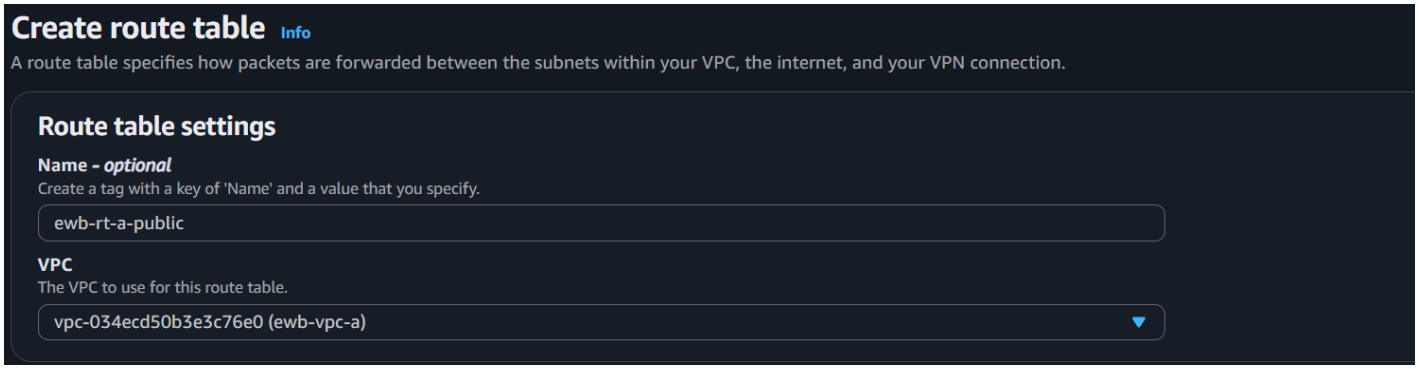




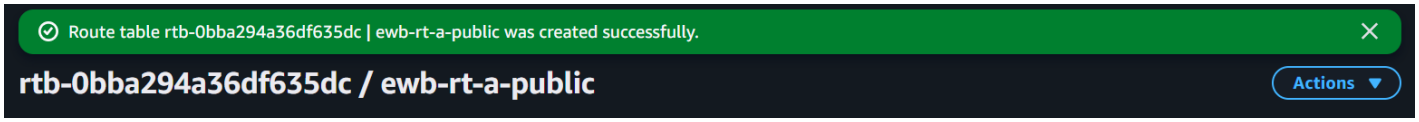


Step 4:- Create the public route table.

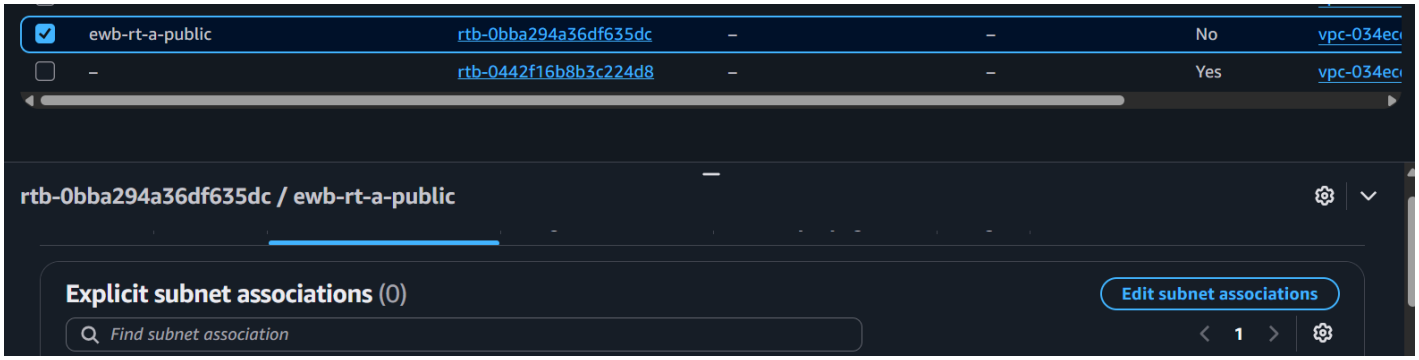


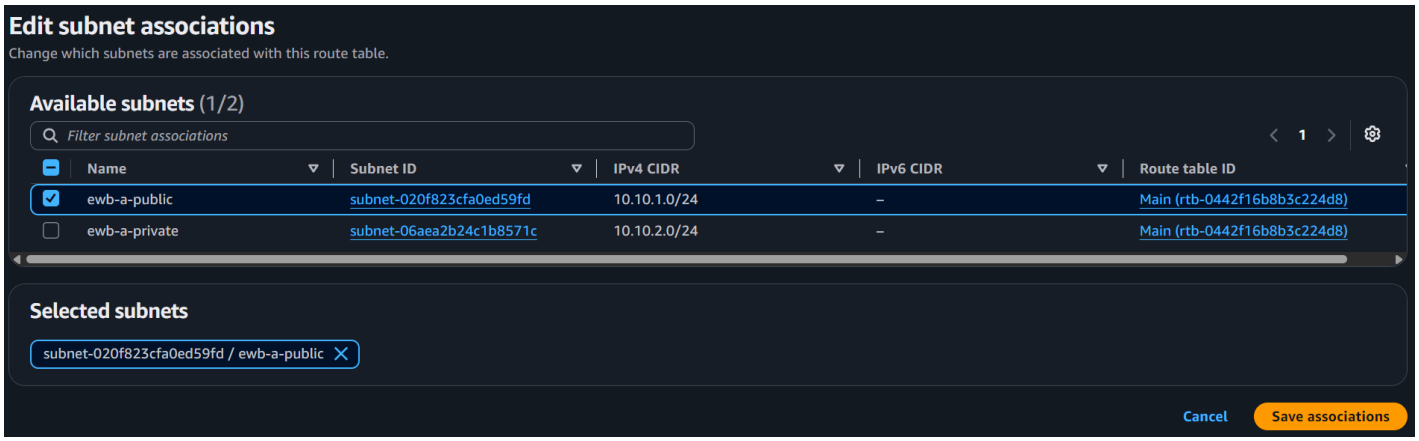


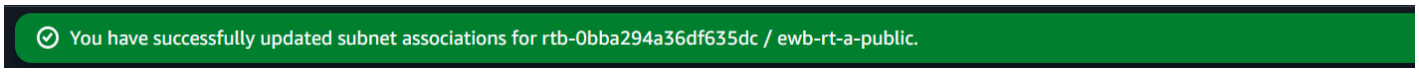




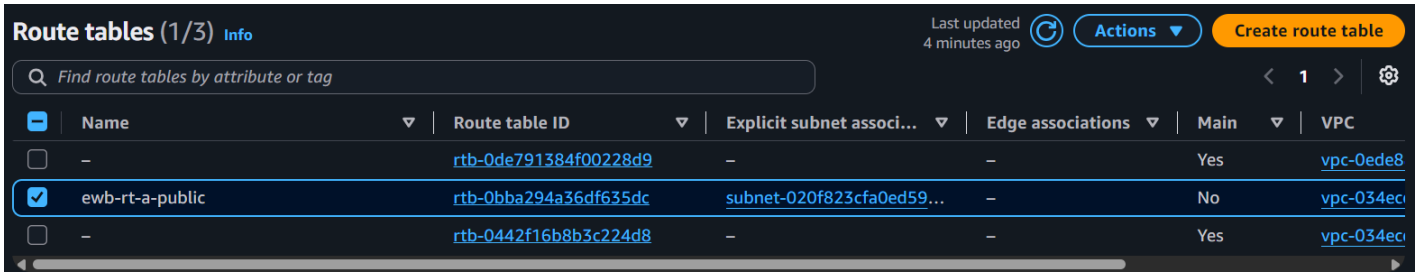
Step 5:- Associate the public subnet with the route table.

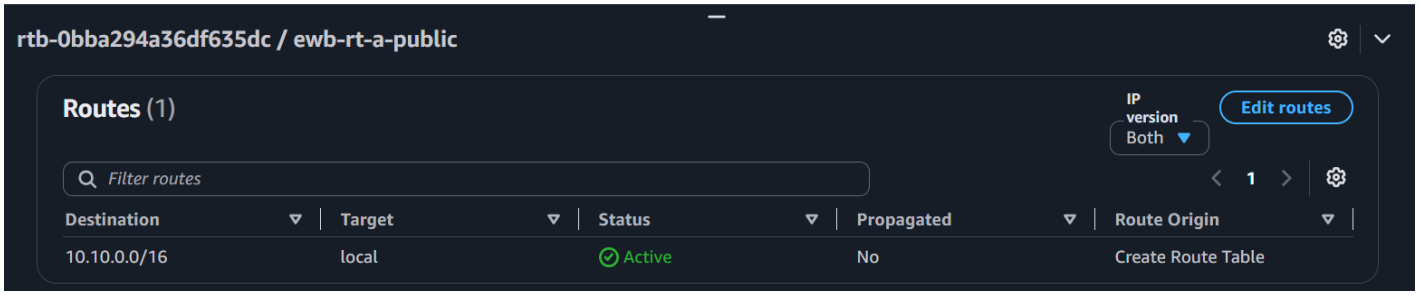


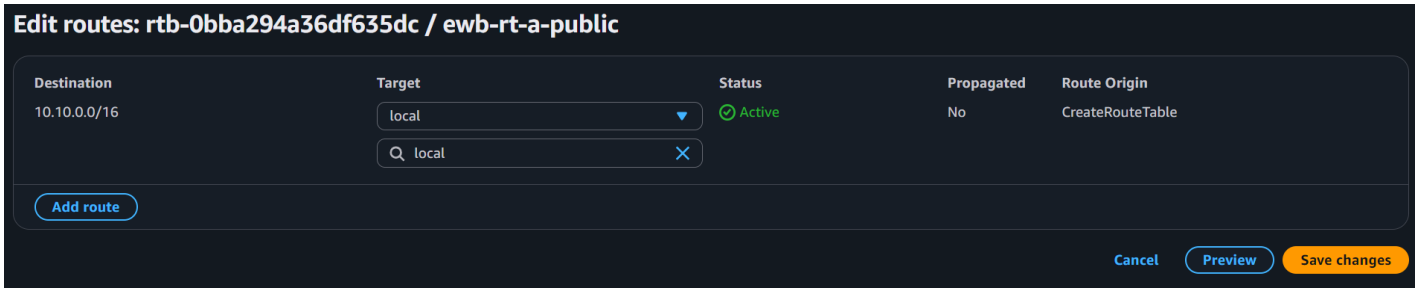


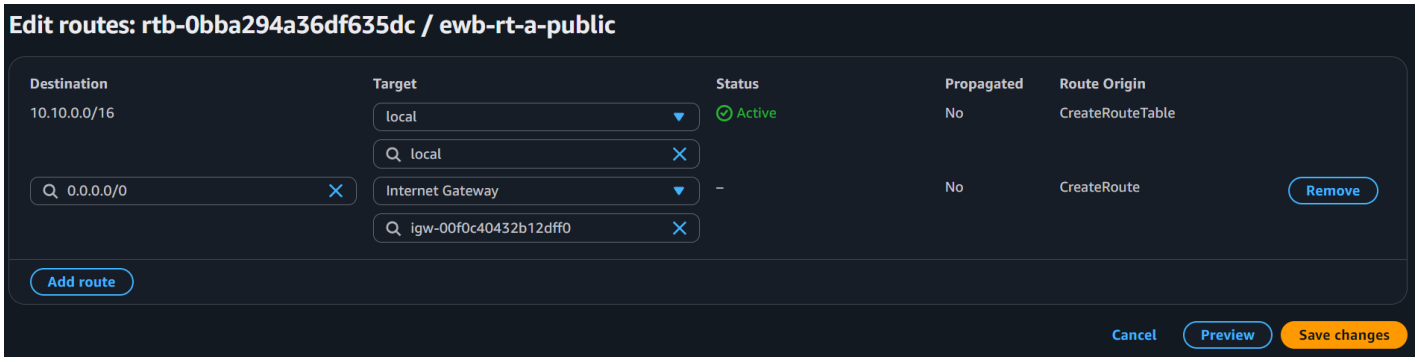


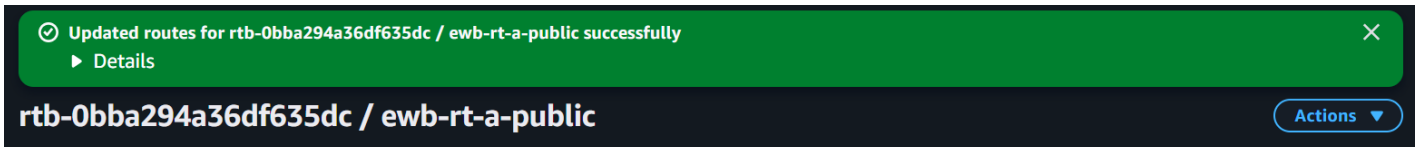
Step 6:- Add the internet route.





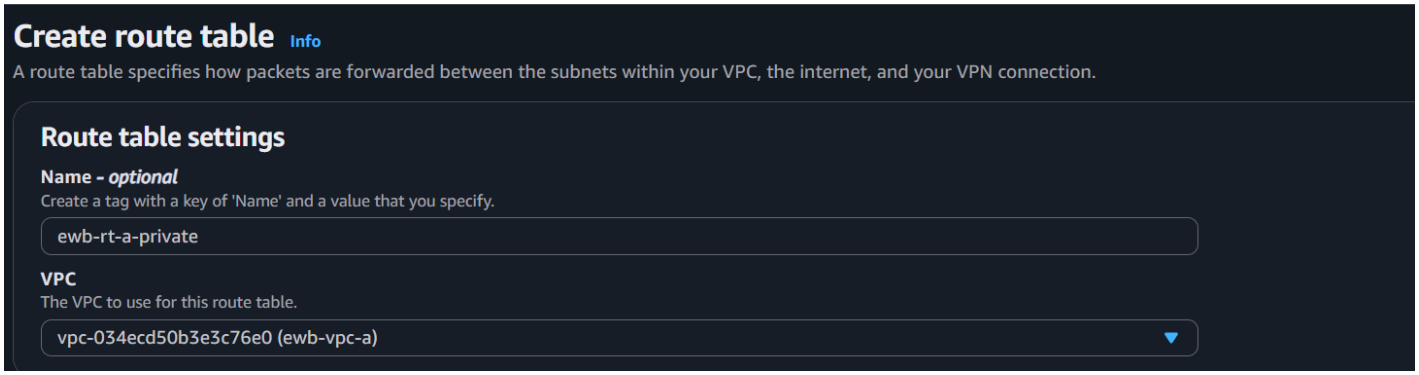






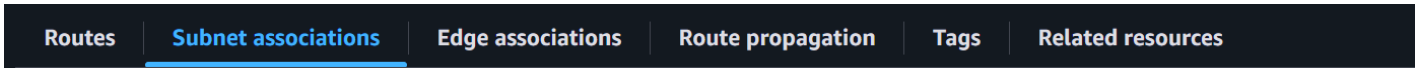
Step 7:- Create the private route table.

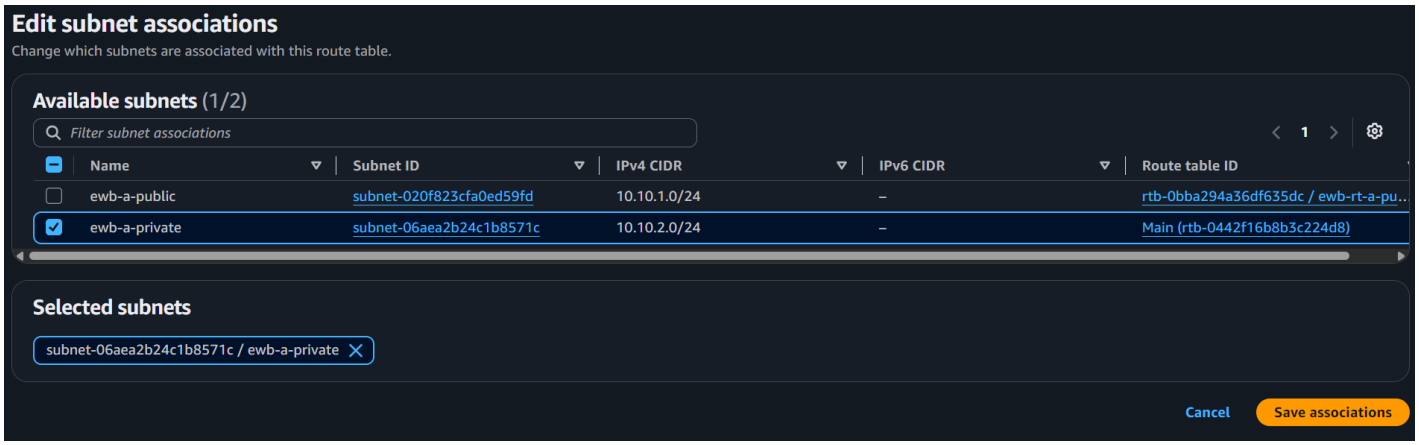






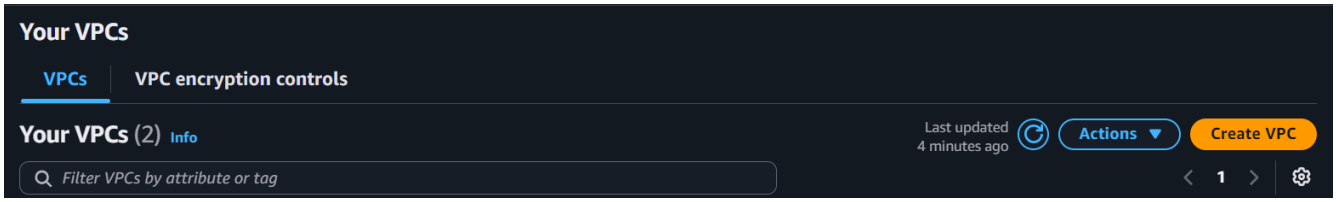
Step 8:- Associate the private subnet.

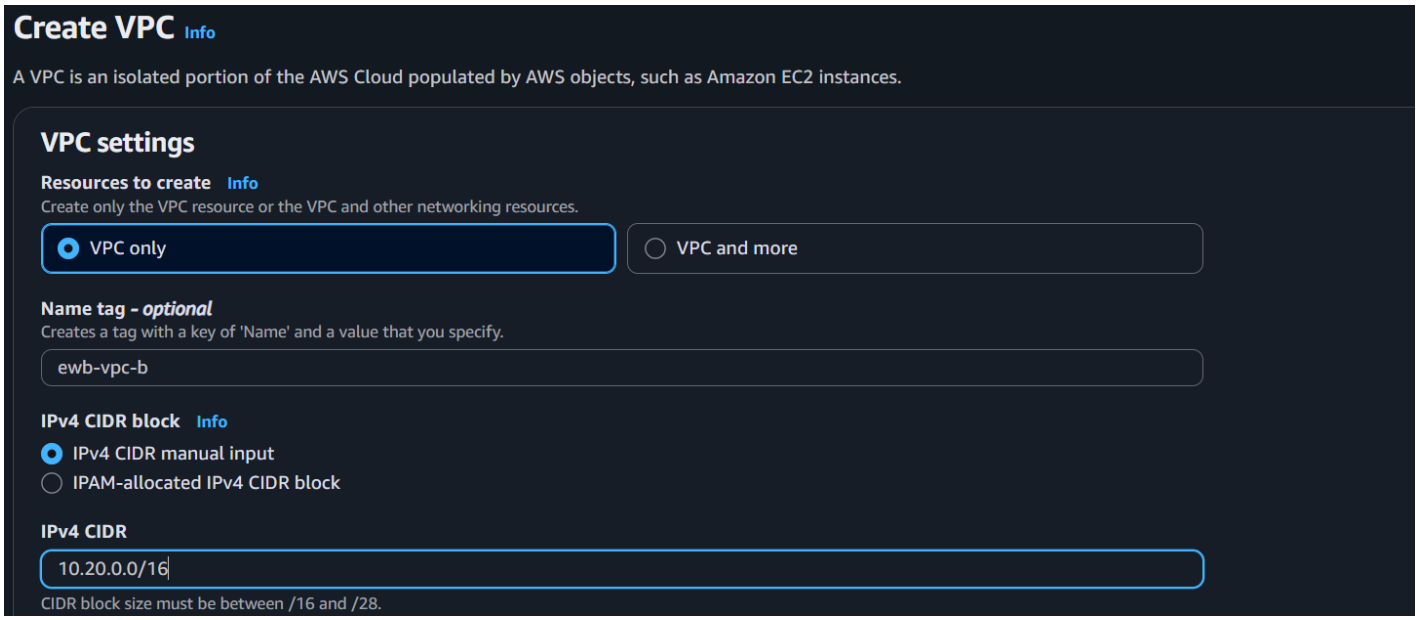


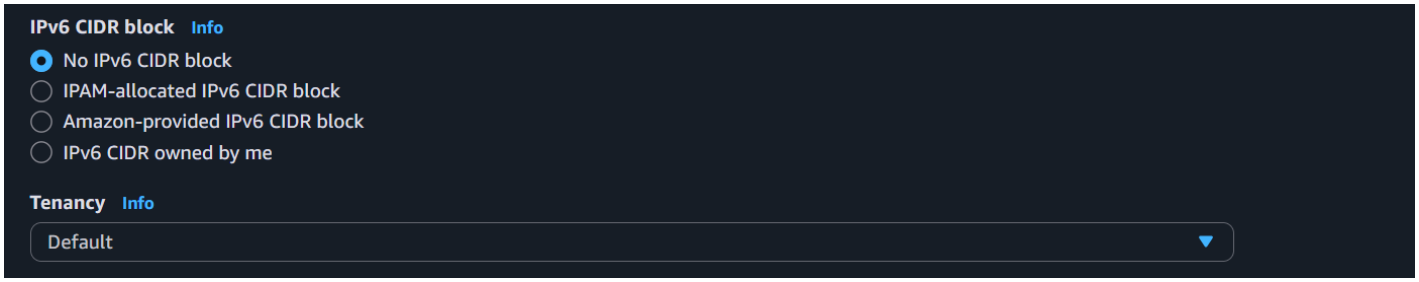


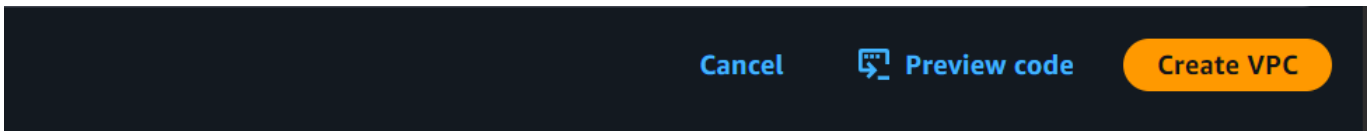


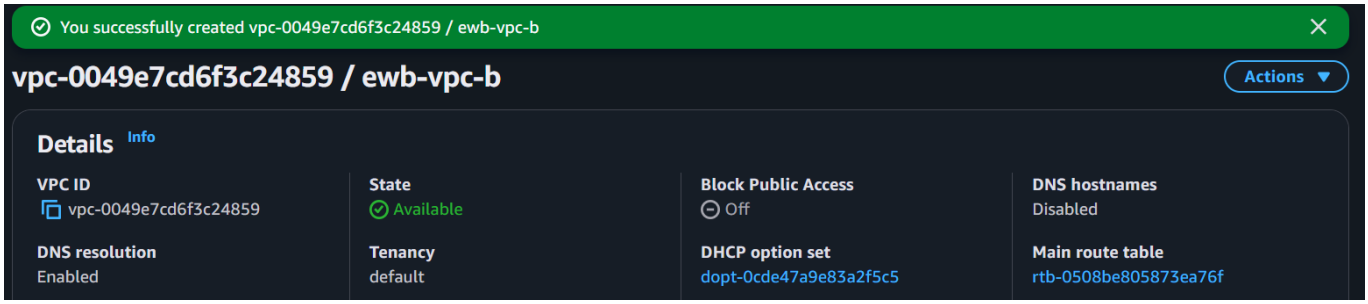
Step 9:- Create VPC B.





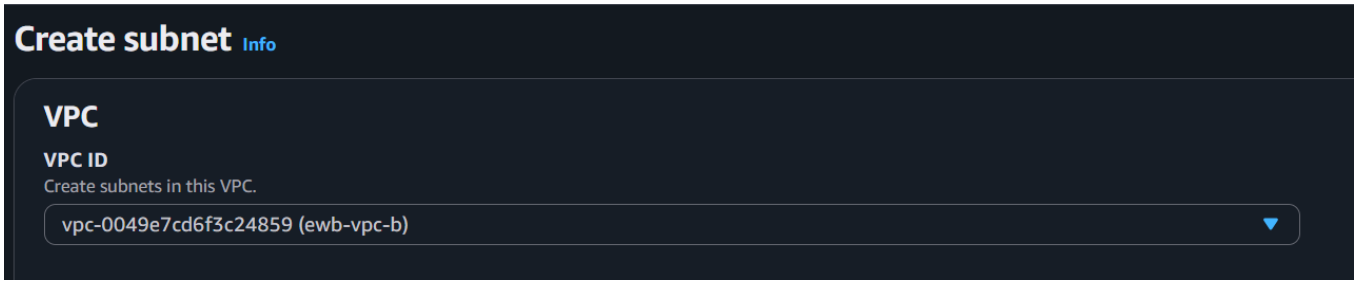


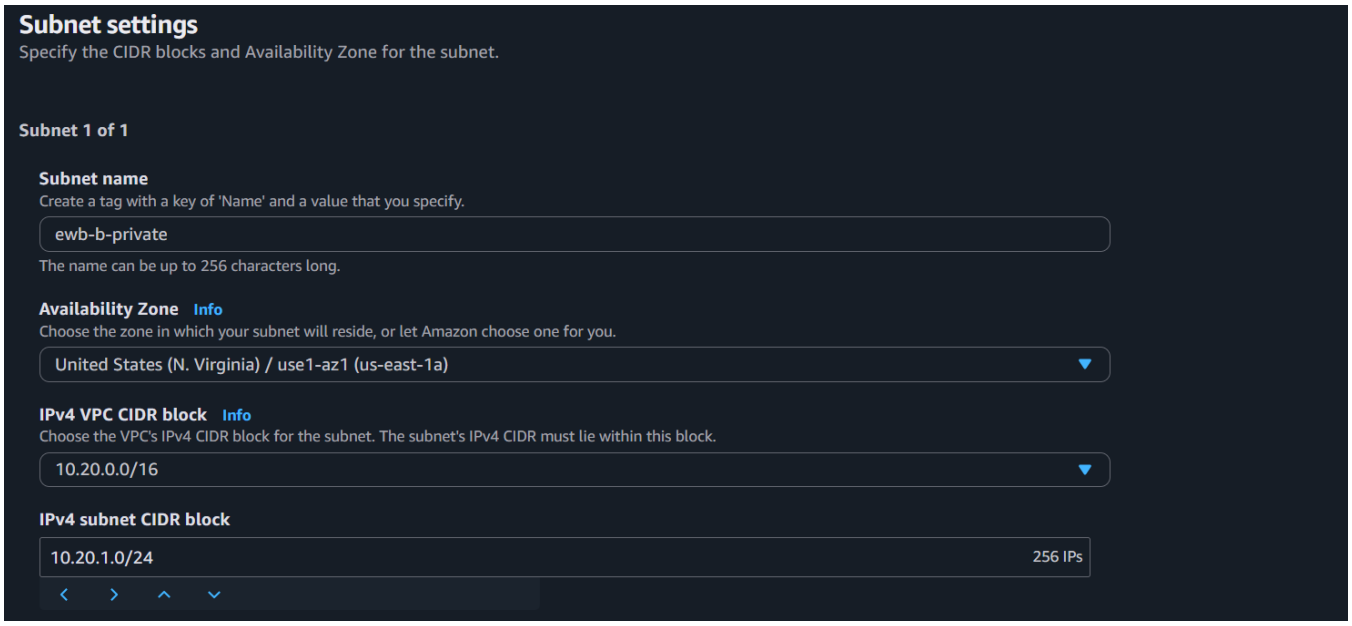




Step 10:- Create a private subnet in VPC B.





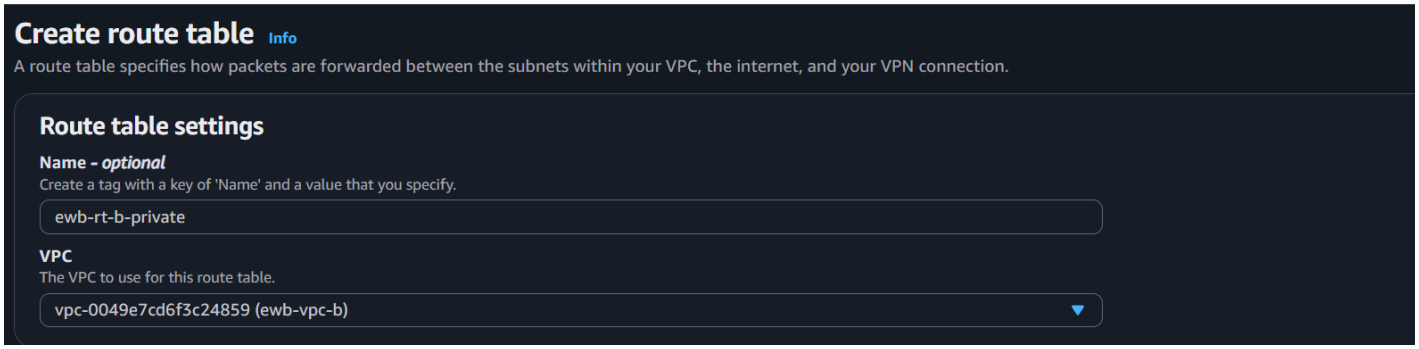




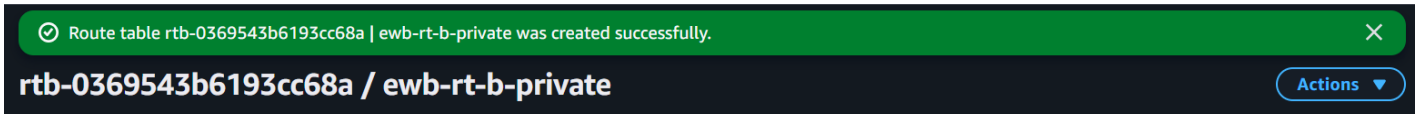


Step 11:- Create VPC B's private route table.

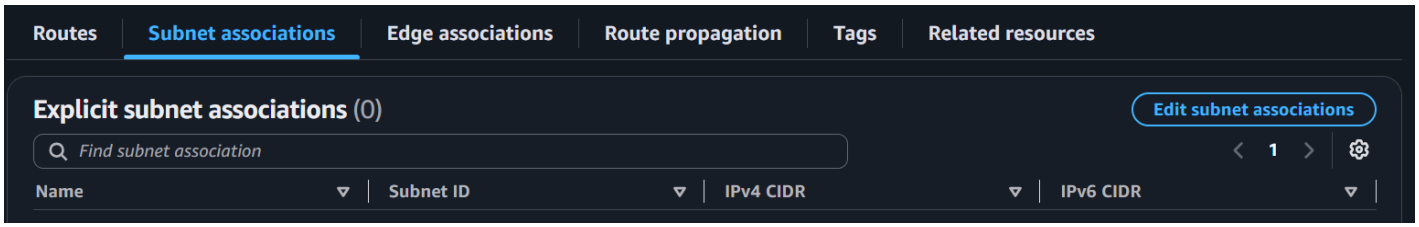


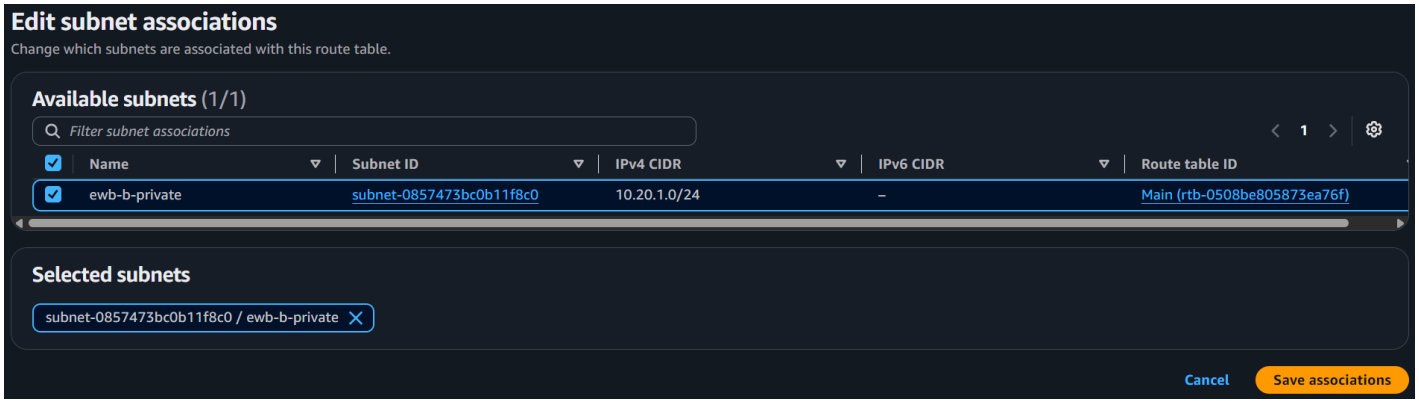




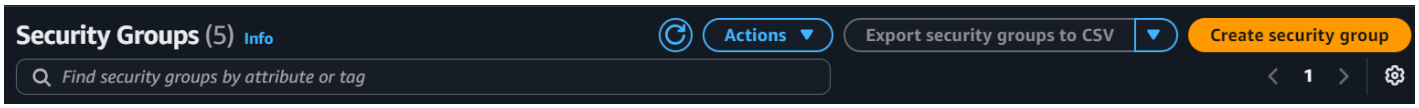


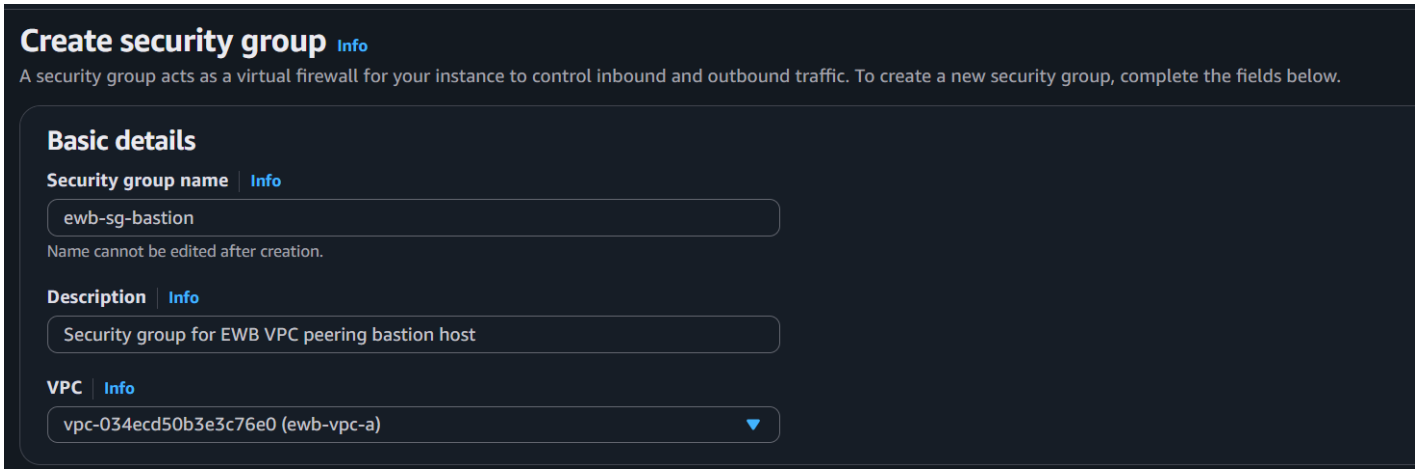
Step 12:- Associate the private subnet with the route table.

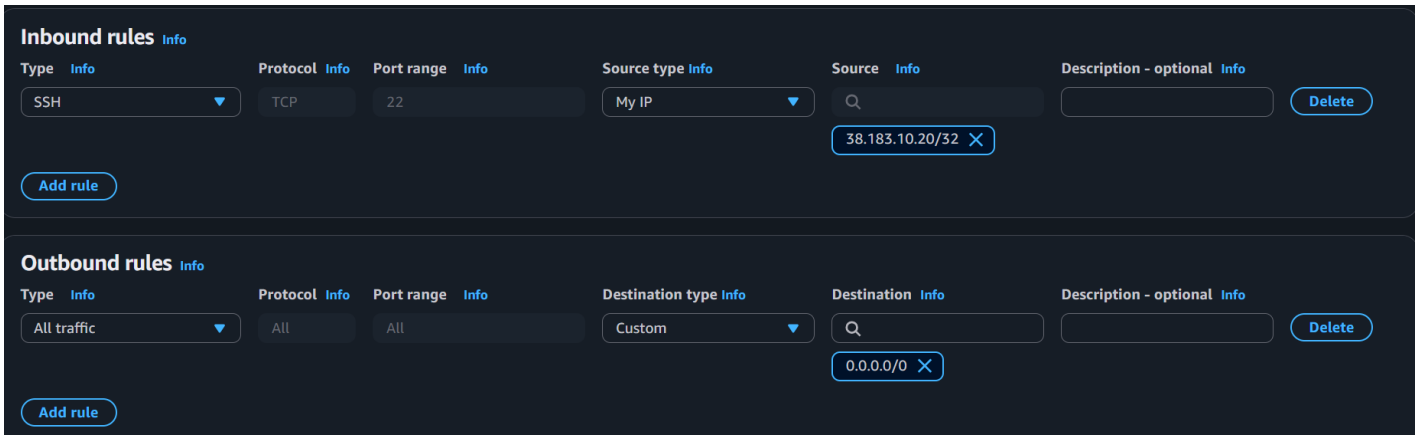


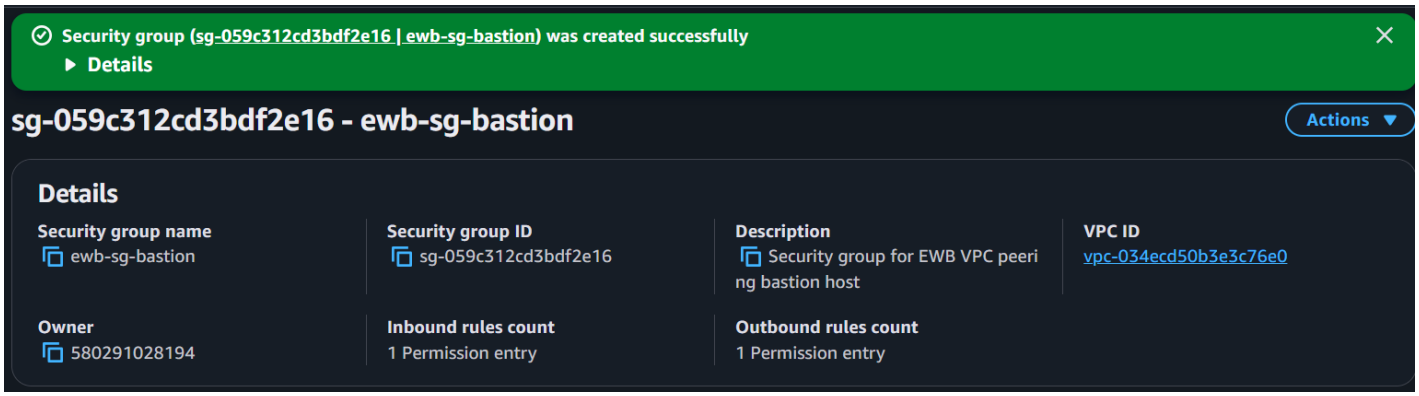


Step 13:- Create a security group for the bastion host.



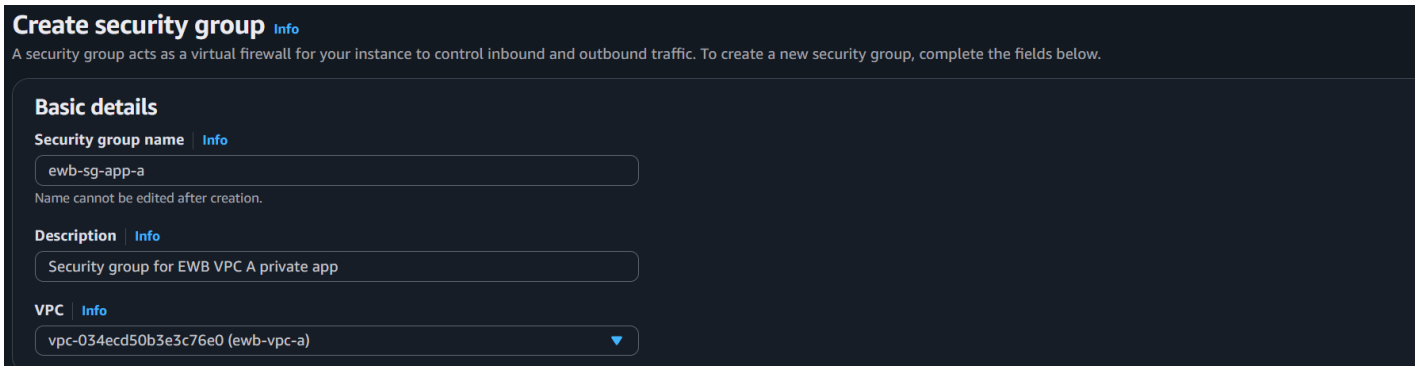


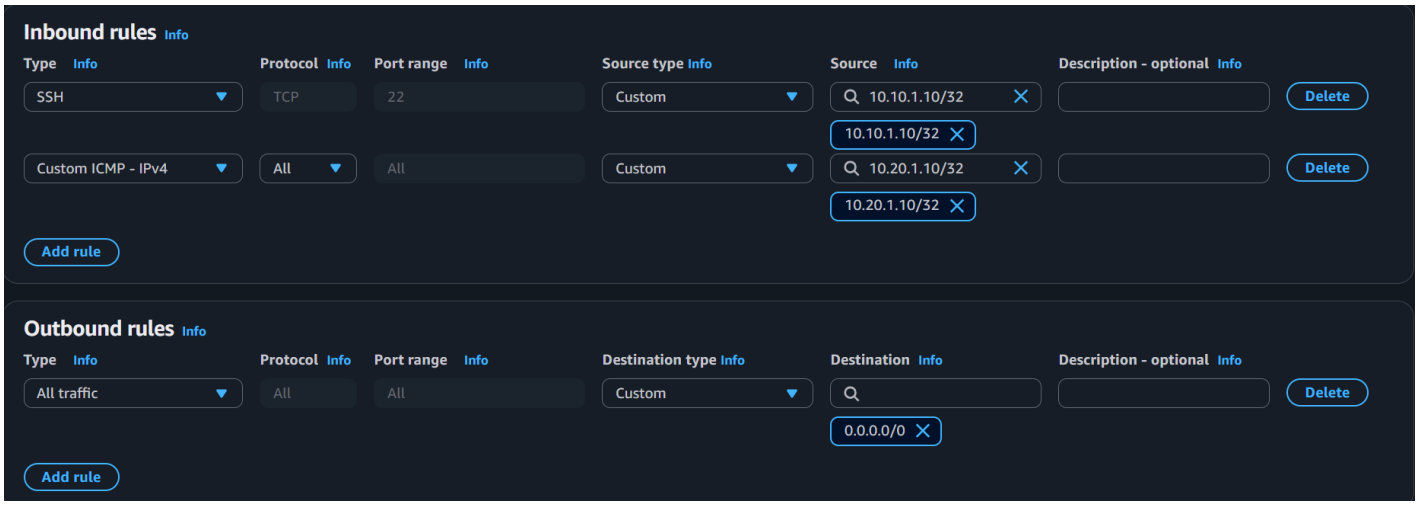




Step 14:- Create the security group for App A.

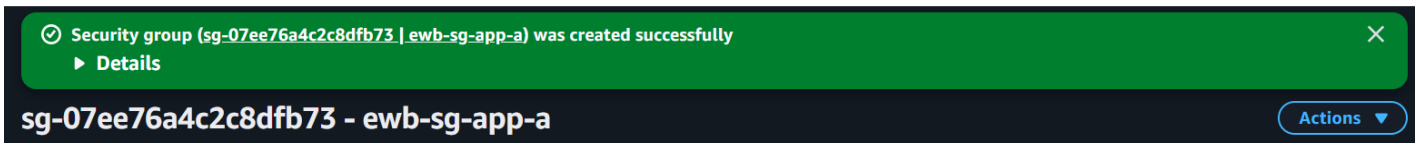




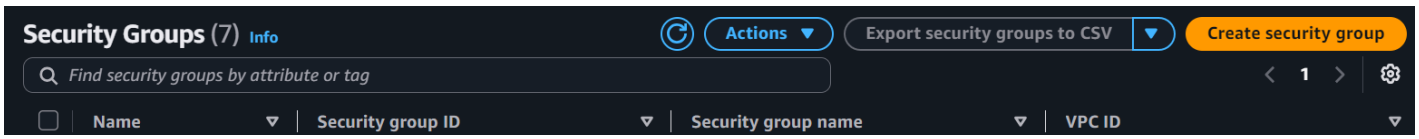


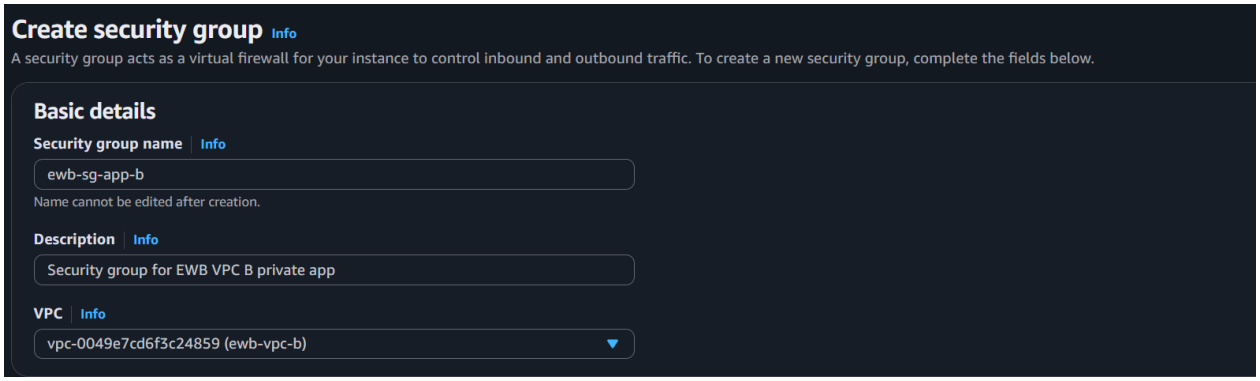


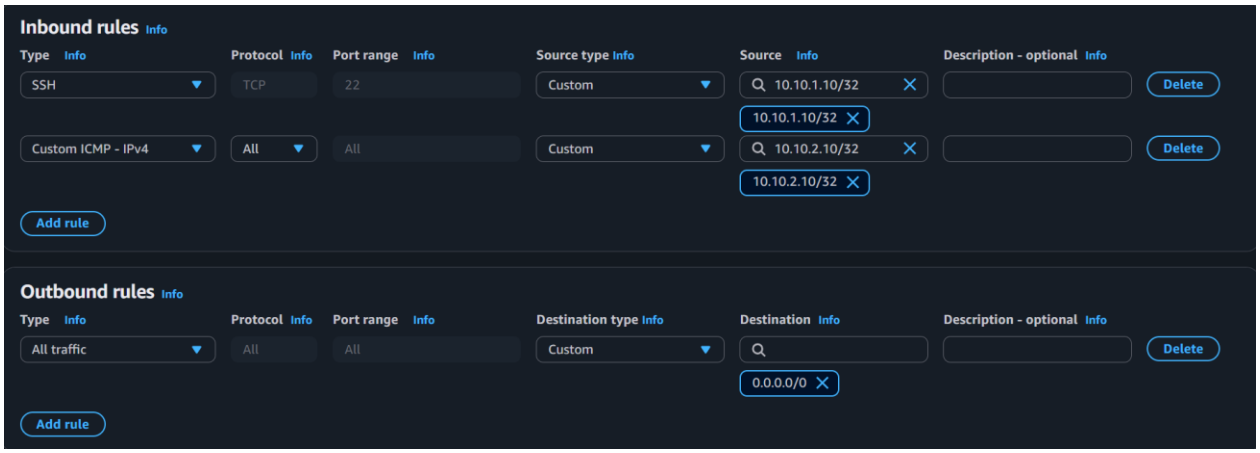




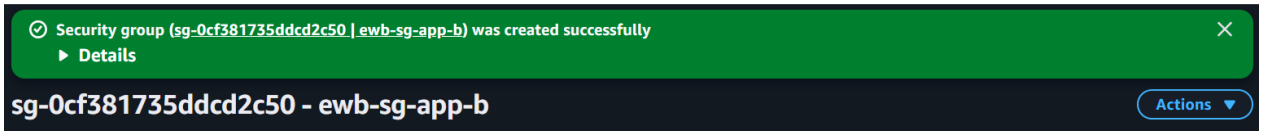
Step 15:- Create the security group for App B.



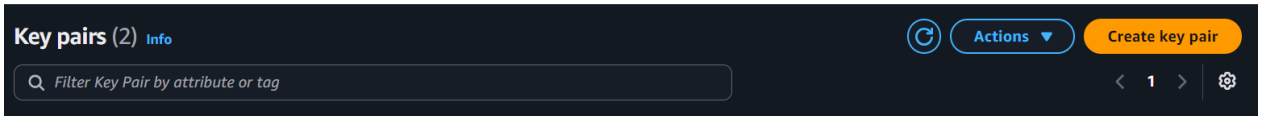


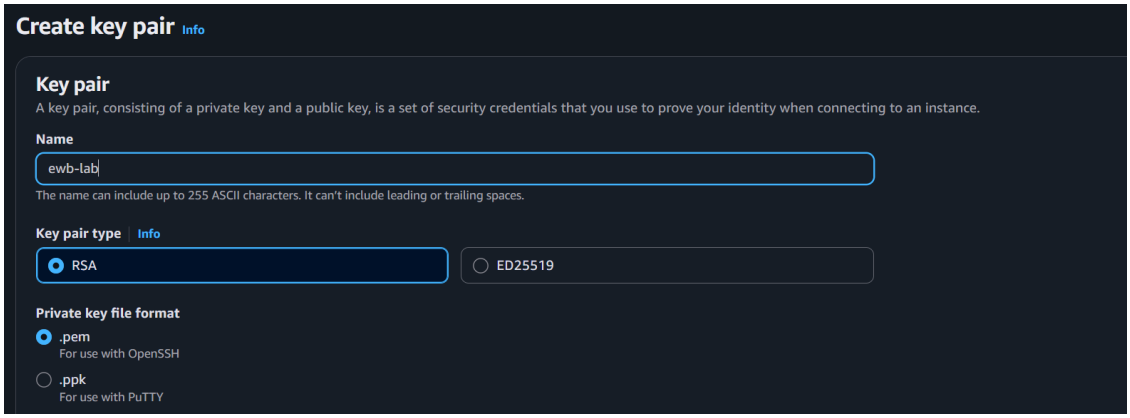






Step 16:- Create an EC2 key pair.

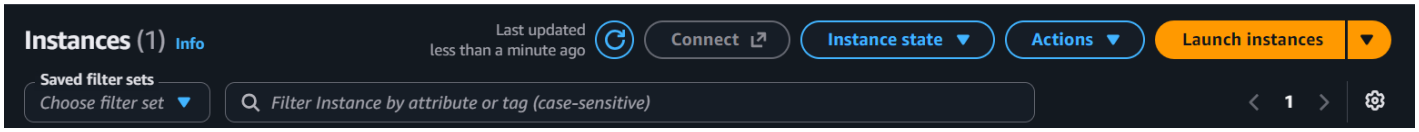


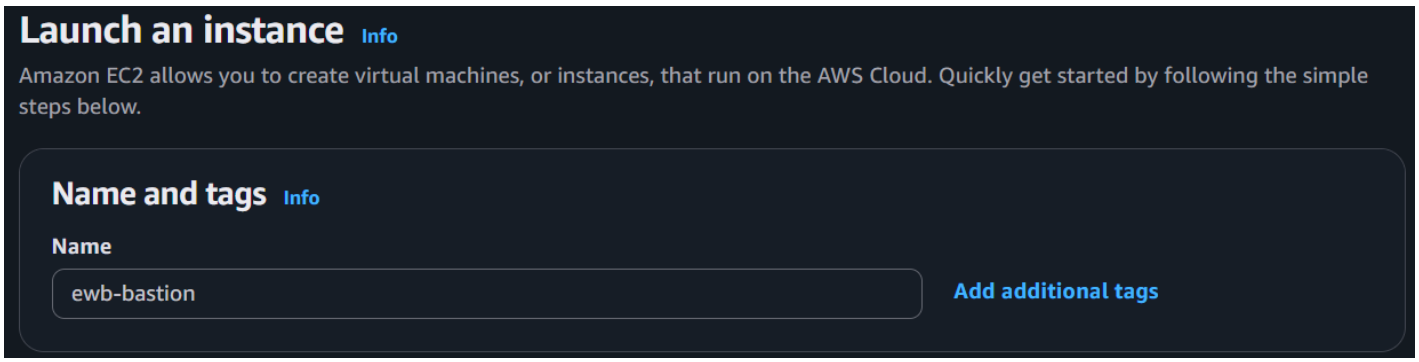


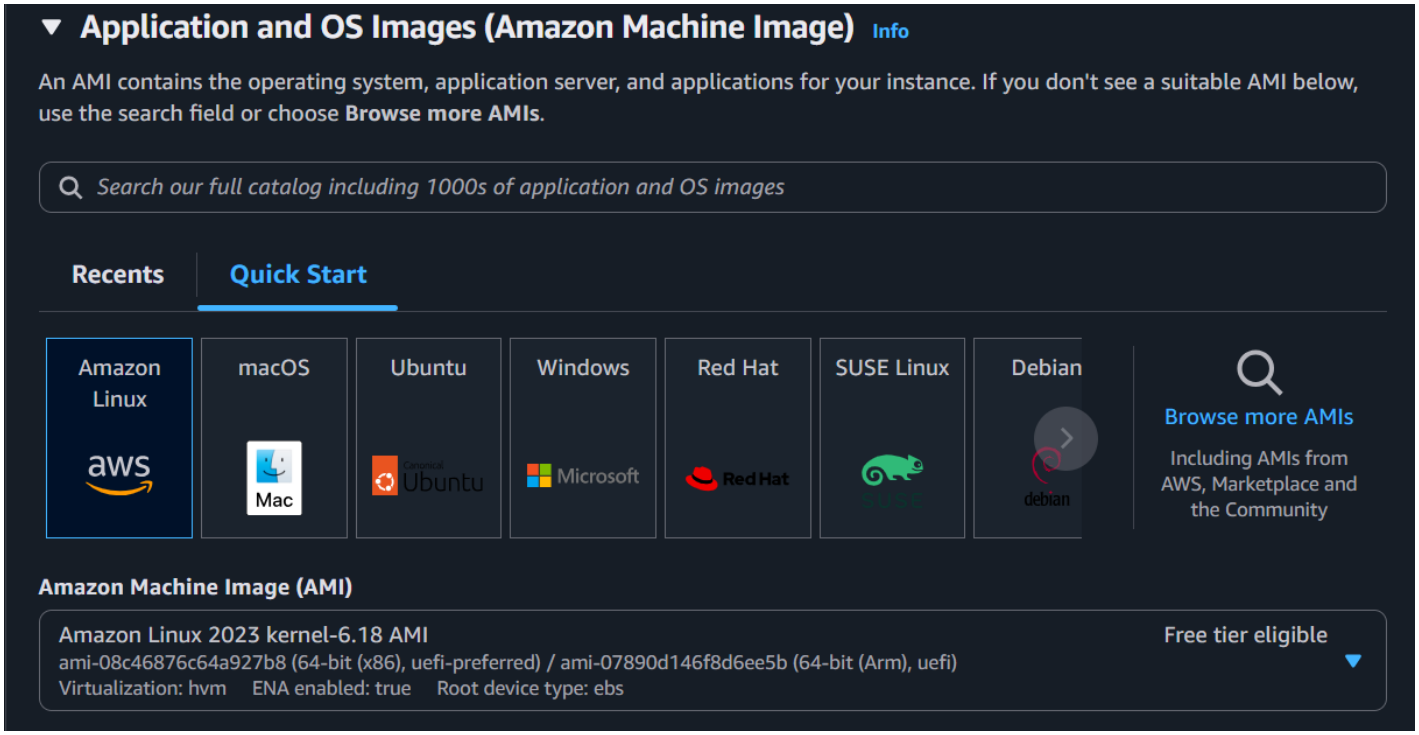




Step 17:- Launch the bastion EC2 instance.







Instance type Info | Get advice

Instance type

t3.micro — Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

ewb-lab

Create new key pair

Network settings Info

VPC - *required* Info

vpc-034ecd50b3e3c76e0 (ewb-vpc-a)
10.10.0.0/16

Subnet Info

subnet-020f823cfa0ed59fd — ewb-a-public
VPC: vpc-034ecd50b3e3c76e0 Owner: 580291028194
Availability Zone: us-east-1a (use1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.10.1.0/24)

Create new subnet

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- ○ Create security group
- ◉ Select existing security group

Common security groups Info

Select security groups

Compare security group rules

ewb-sg-bastion sg-059c312cd3bdf2e16 ✕
VPC: vpc-034ecd50b3e3c76e0

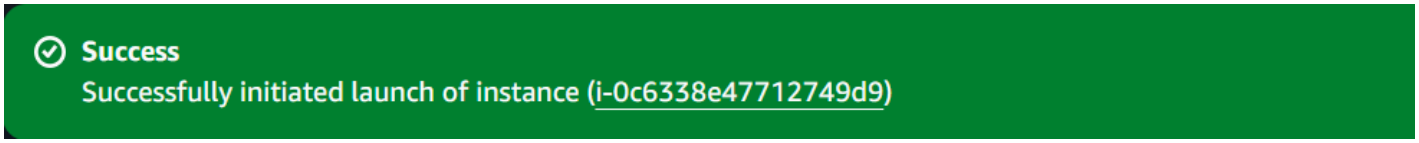
Security groups that you add or remove here will be added to or removed from all your network interfaces.

Cancel

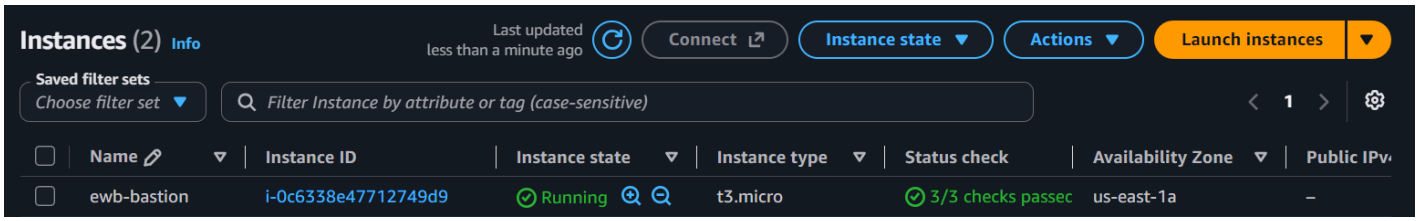
Launch instance

Preview code

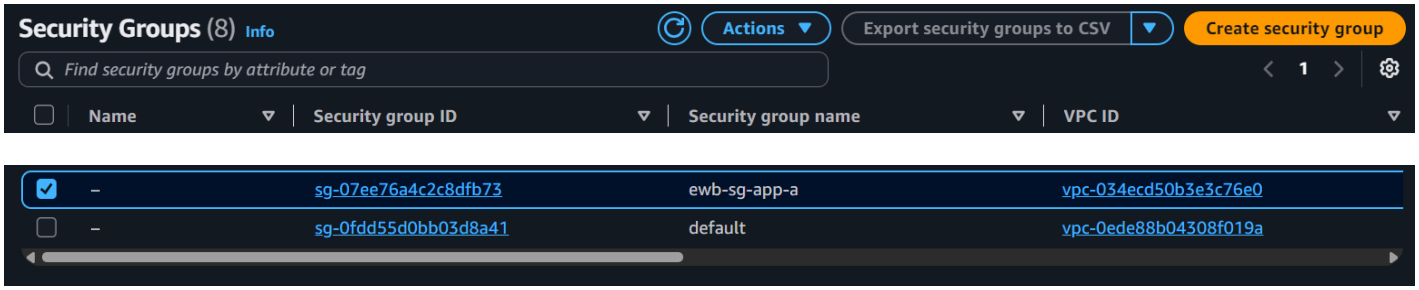


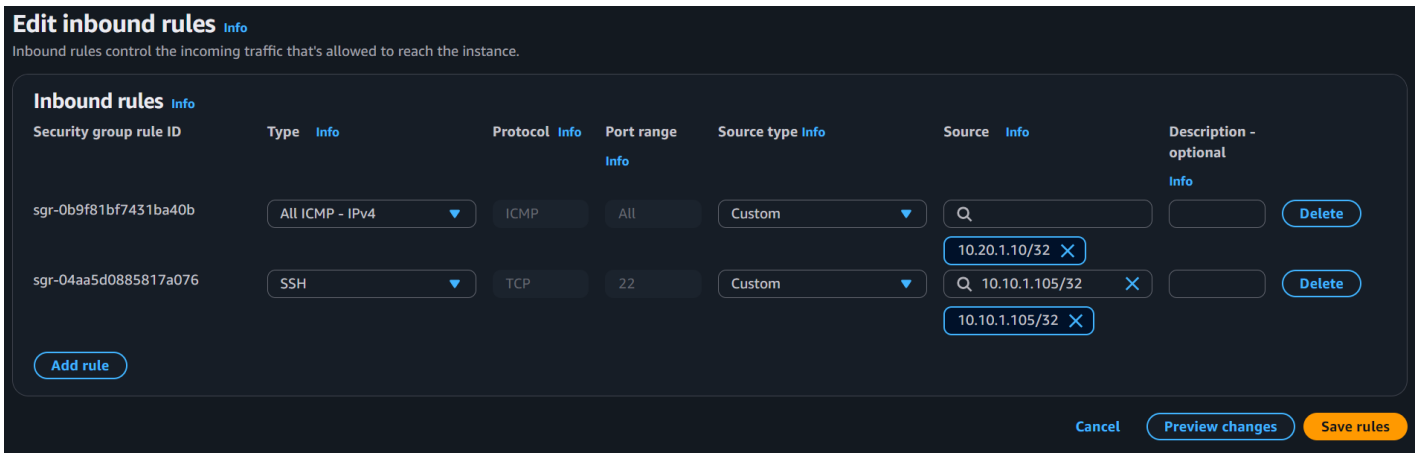


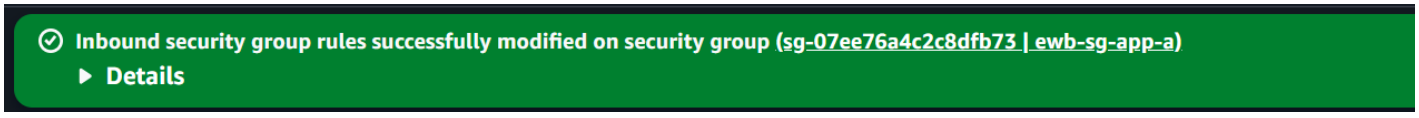
Step 18:- Verify the bastion instance.



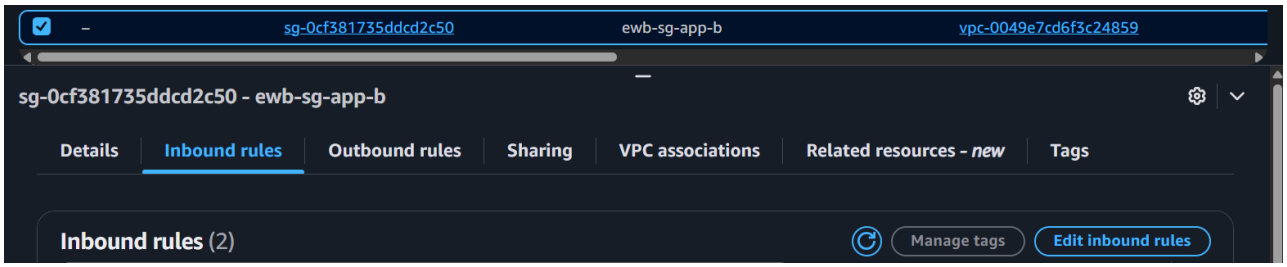
Step 19:- Update App A's security group.

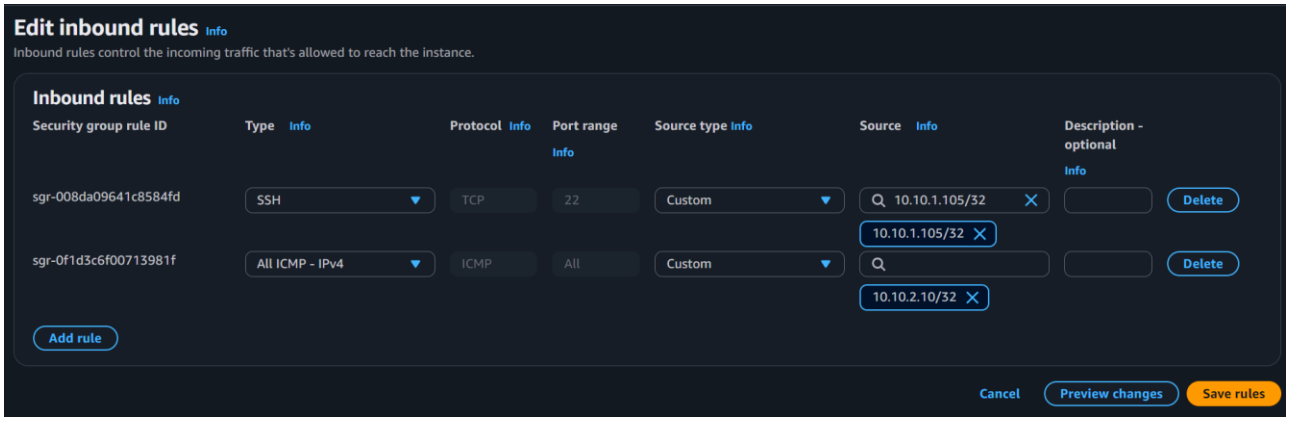


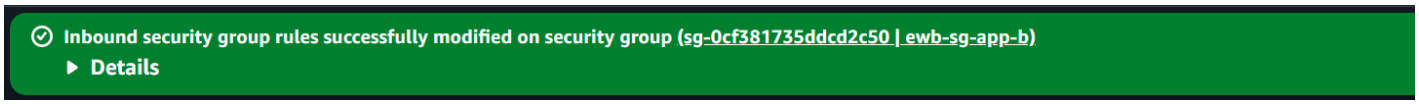




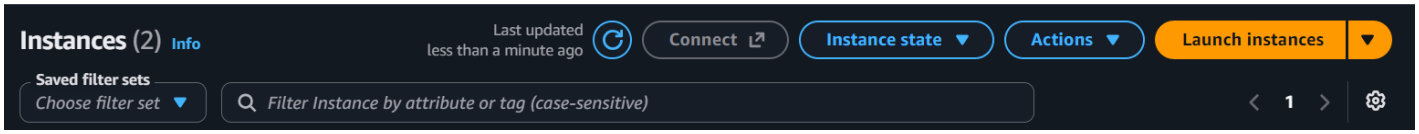
Step 20:- Update App B's security group.

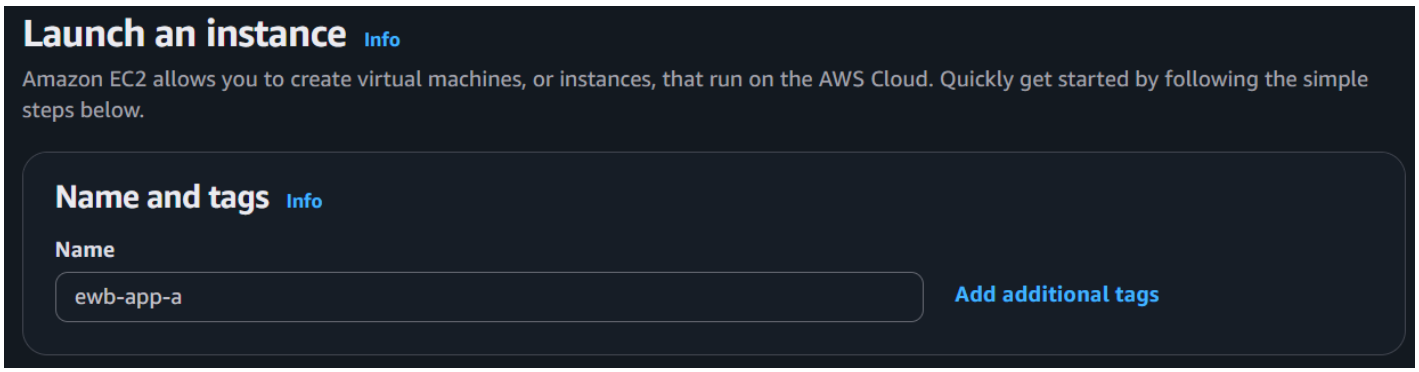


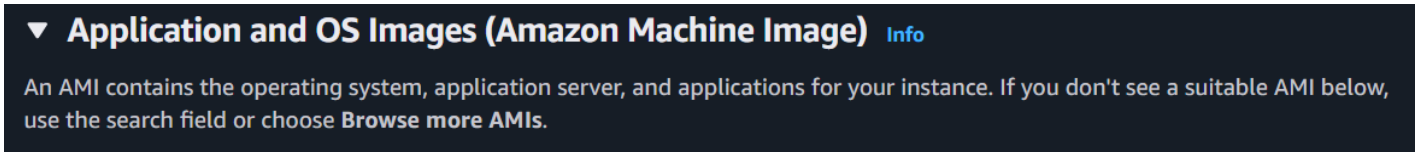


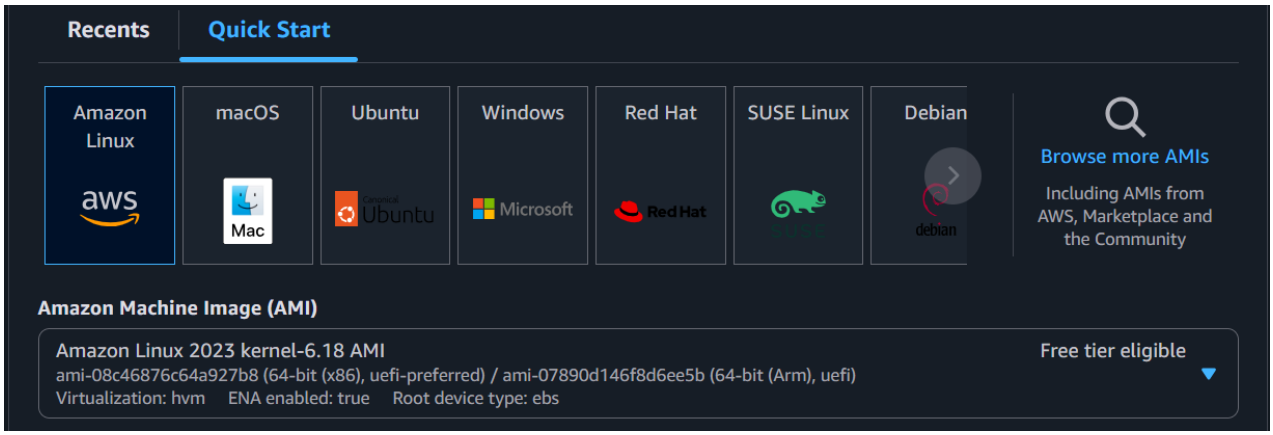


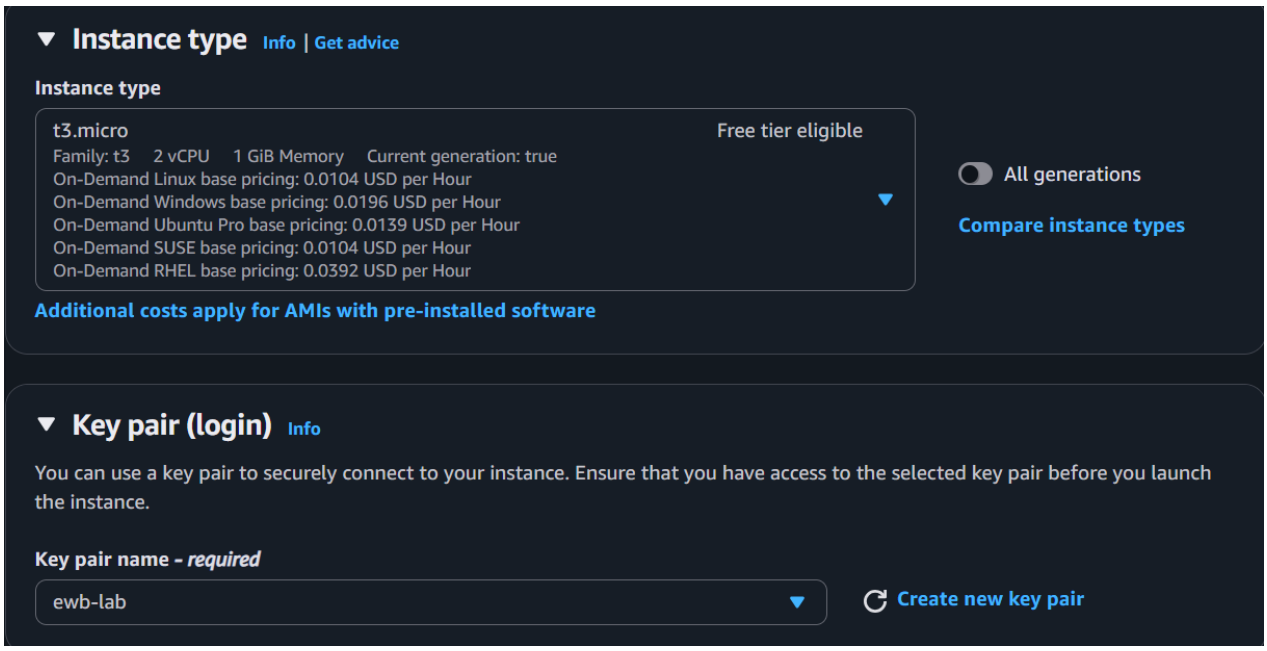
Step 21:- Launch the App A EC2 instance.



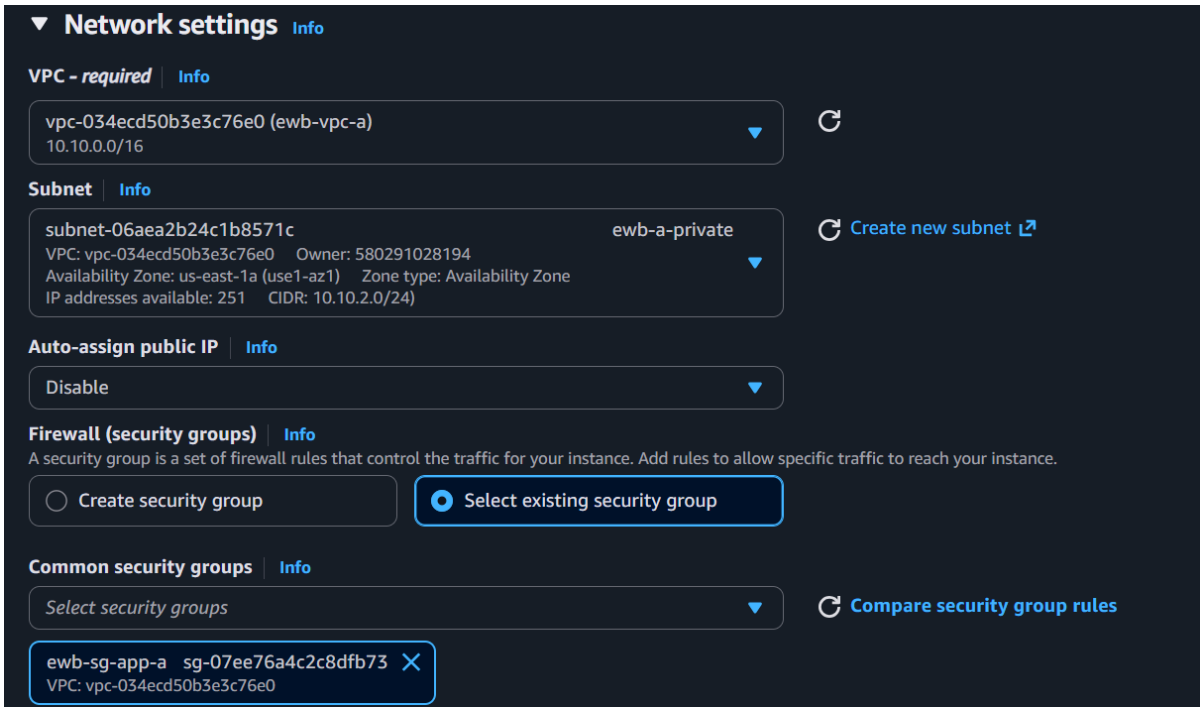





Recents
Quick Start
Amazon Linux
aws
macOS
Mac
Ubuntu
Ubuntu
Windows
Microsoft
Red Hat
Red Hat
SUSE Linux
Debian
debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
ami-08c46876c64a927b8 (64-bit (x86), uefi-preferred) / ami-07890d146f8d6ee5b (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs


Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software

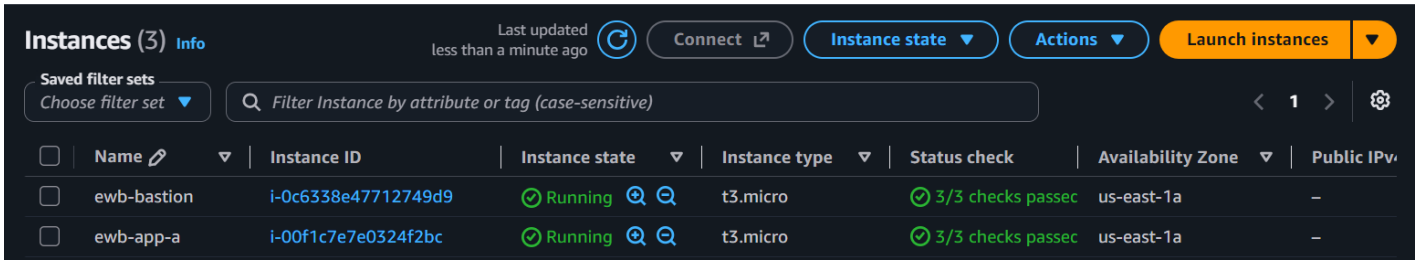
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
ewb-lab
Create new key pair


Network settings Info
VPC - required Info
vpc-034ecd50b3e3c76e0 (ewb-vpc-a)
10.10.0.0/16
Subnet Info
subnet-06aea2b24c1b8571c
ewb-a-private
VPC: vpc-034ecd50b3e3c76e0 Owner: 580291028194
Availability Zone: us-east-1a (use1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.10.2.0/24)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
ewb-sg-app-a sg-07ee76a4c2c8dfb73
VPC: vpc-034ecd50b3e3c76e0

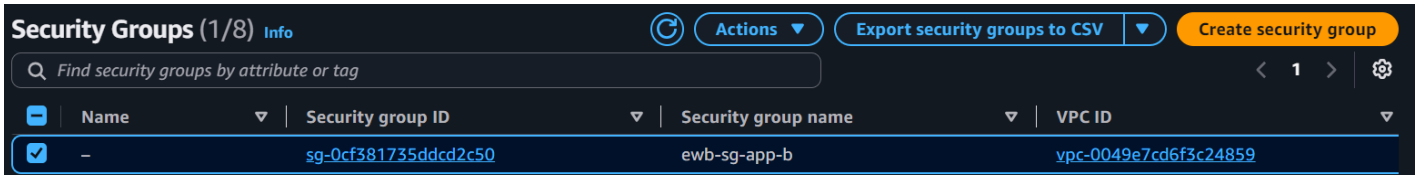


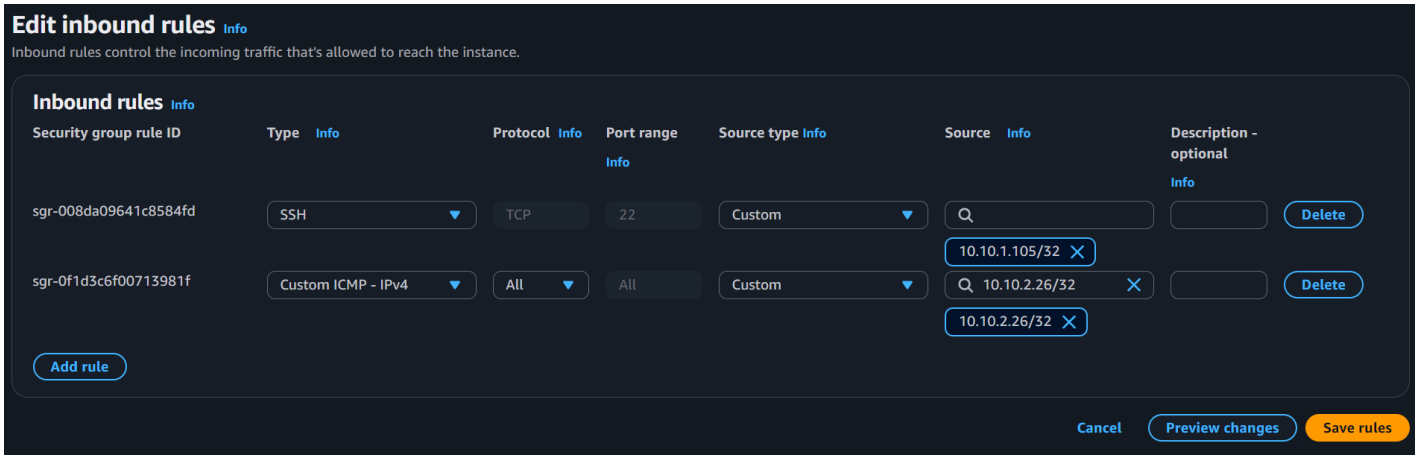


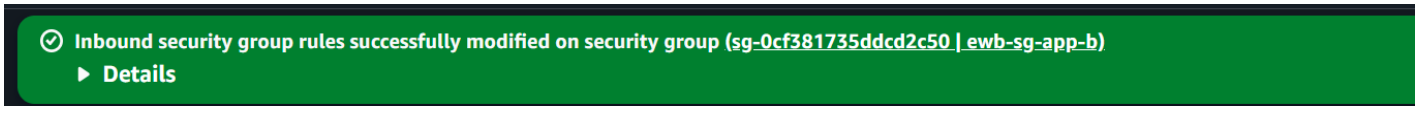
Step 22:- Verify App A.



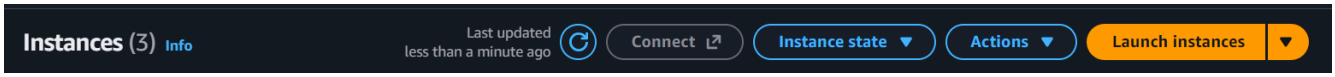
Step 23:- Update App B's security group.







Step 24:- Launch the App B EC2 instance.



Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

ewb-app-b

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI — Free tier eligible

ami-08c46876c64a927b8 (64-bit (x86), uefi-preferred) / ami-07890d146f8d6ee5b (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Instance type Info | Get advice

Instance type

t3.micro — Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

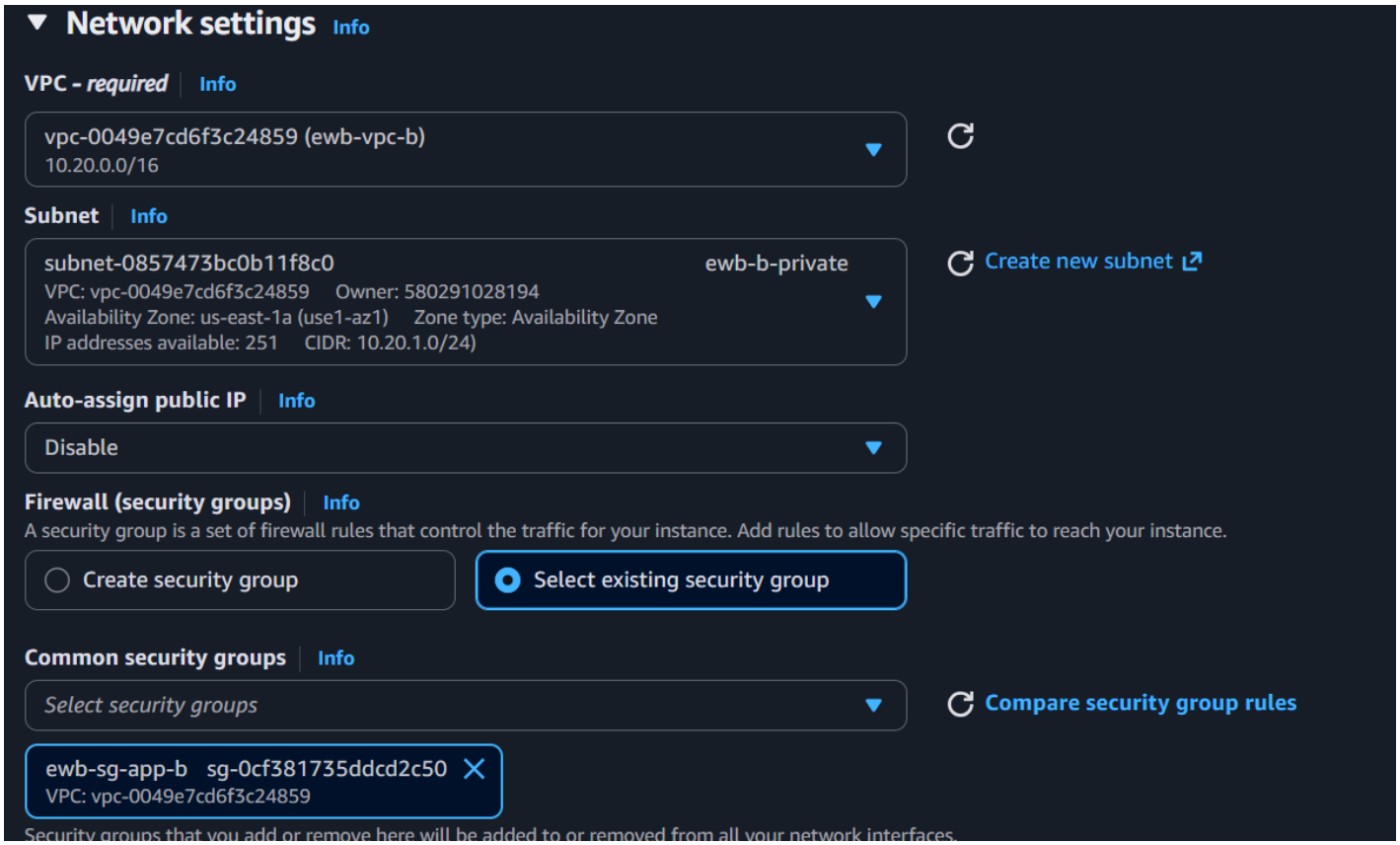
Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

ewb-lab

Create new key pair

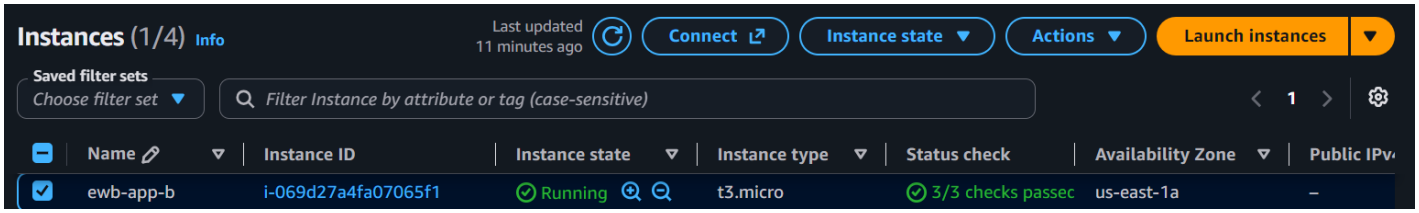




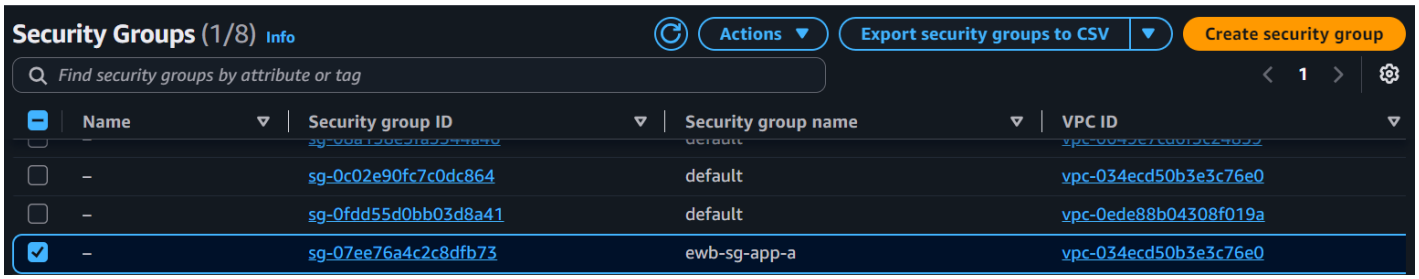


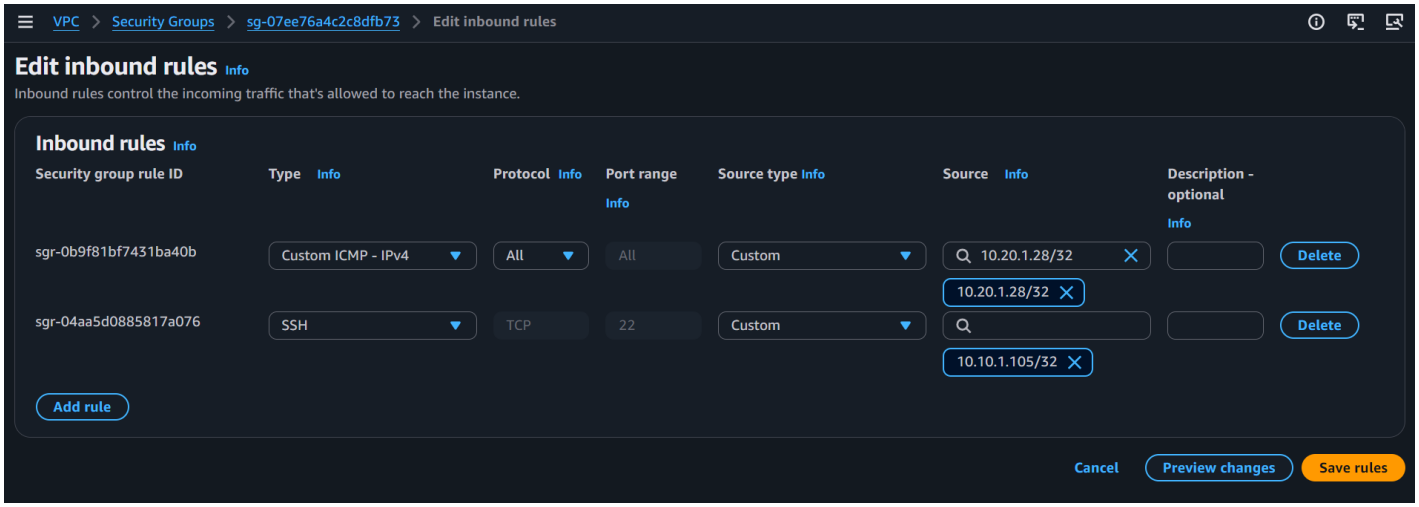


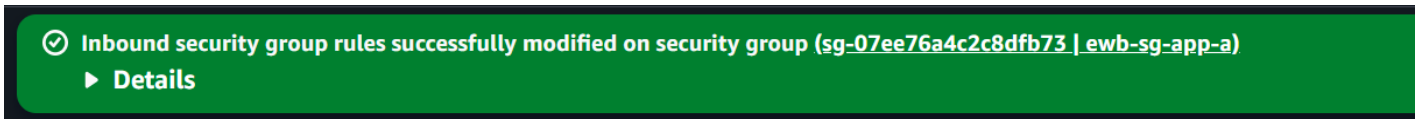
Step 25:- Verify App B's private IP.



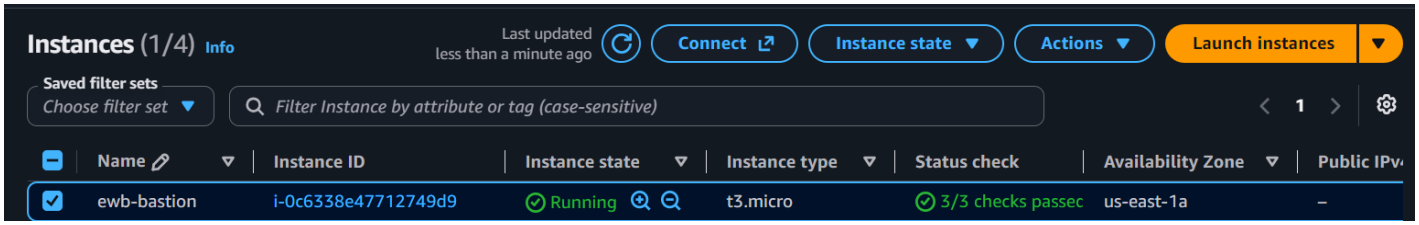
Step 26:- Update App A's ICMP security rule.

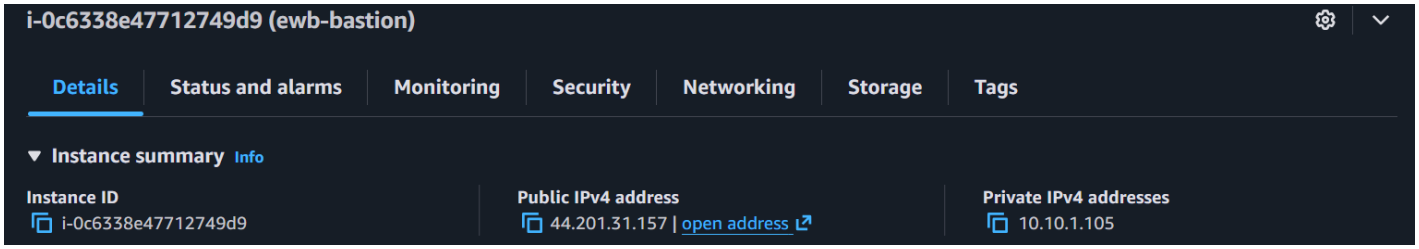




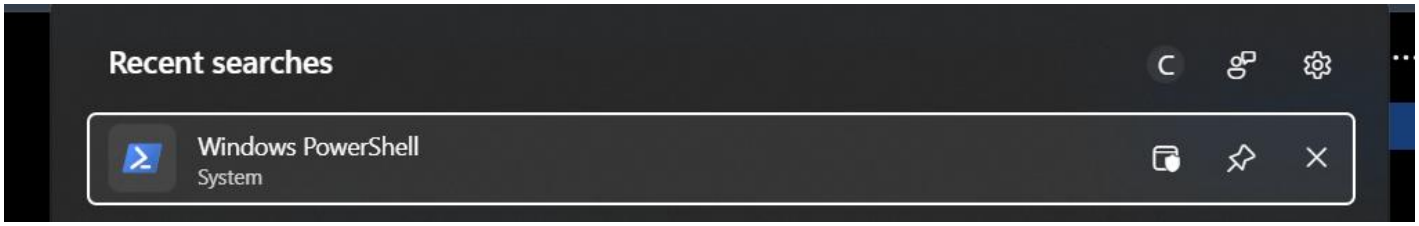


Step 27:- Test connectivity before VPC peering.



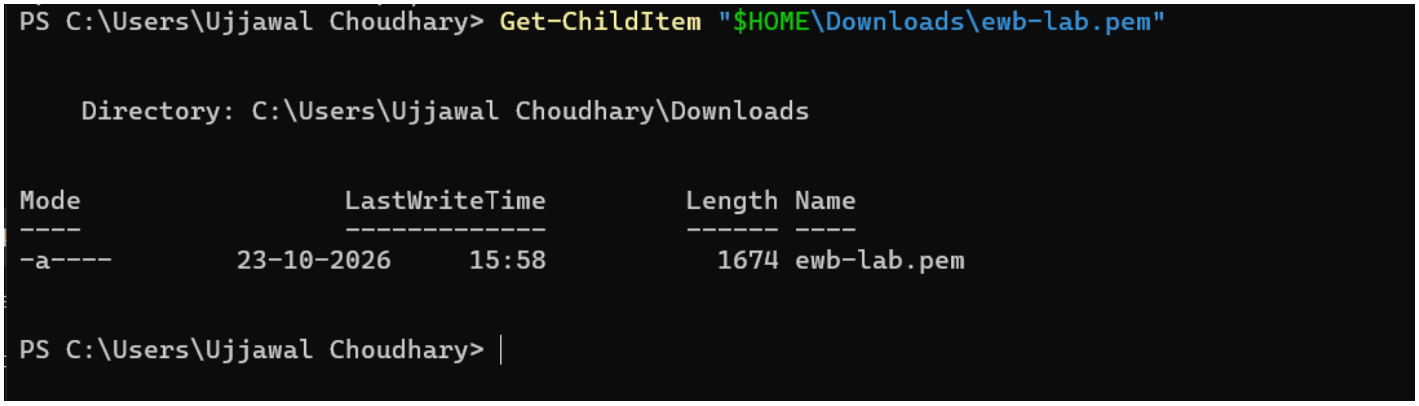


Step 28:- Prepare your SSH key.

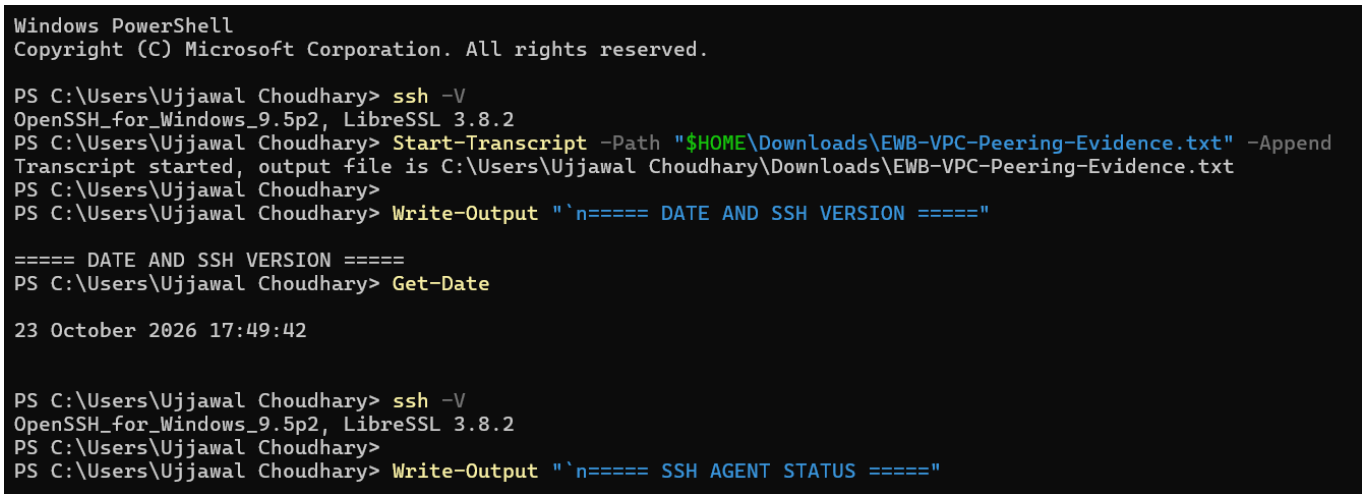




Step 29:- Locate your SSH key.



Step 30:- Connect to the bastion with agent forwarding.



```
PS C:\Users\Ujjawal Choudhary>
PS C:\Users\Ujjawal Choudhary> Write-Output "`n===== SSH KEYS LOADED ====="

===== SSH KEYS LOADED =====
PS C:\Users\Ujjawal Choudhary> ssh-add -l
2048 SHA256:ssrU03ysvoco9yLBZ2x0jz2AV5RJOtTiUgWTD99ygqE C:\Users\Ujjawal Choudhary\Downloads\ewb-lab.pem (RSA)
PS C:\Users\Ujjawal Choudhary>
PS C:\Users\Ujjawal Choudhary> Write-Output "`n===== CONNECT TO BASTION ====="

===== CONNECT TO BASTION =====
PS C:\Users\Ujjawal Choudhary> ssh -A -i "$HOME\Downloads\ewb-lab.pem" ec2-user@44.201.31.157
```

Step 31:- Connect to the bastion again.

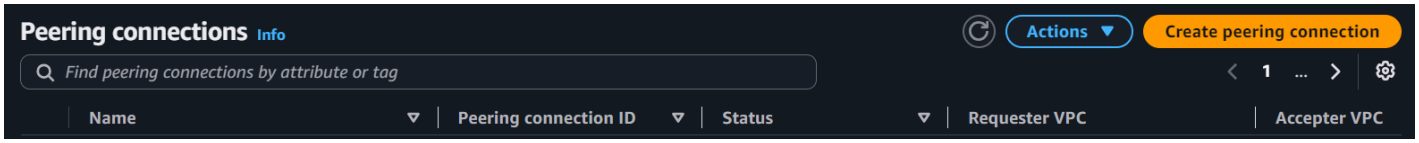
```
PS C:\Users\Ujjawal Choudhary> ssh -A -i "$HOME\Downloads\ewb-lab.pem" ec2-user@44.201.31.157
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 12:14:52 2026 from 38.183.10.20
[ec2-user@ip-10-10-1-105 ~]$ ssh ec2-user@10.10.2.26
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-26 ~]$
```

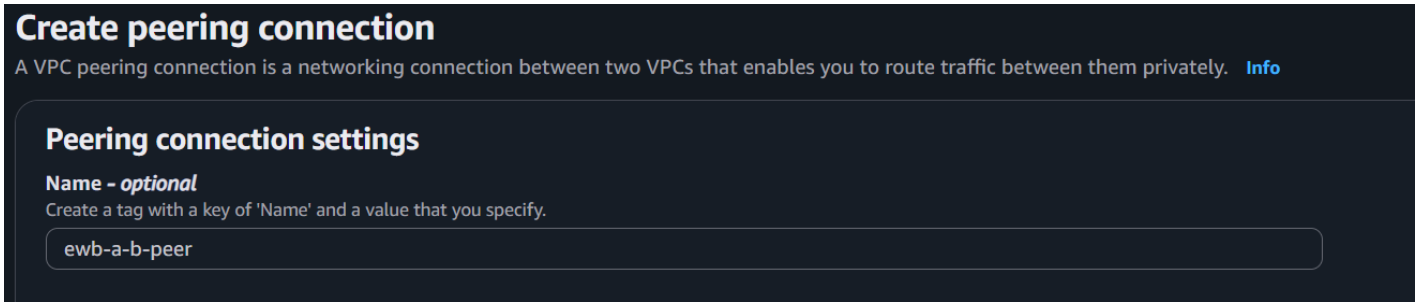
Step 32:- Test the baseline connection.

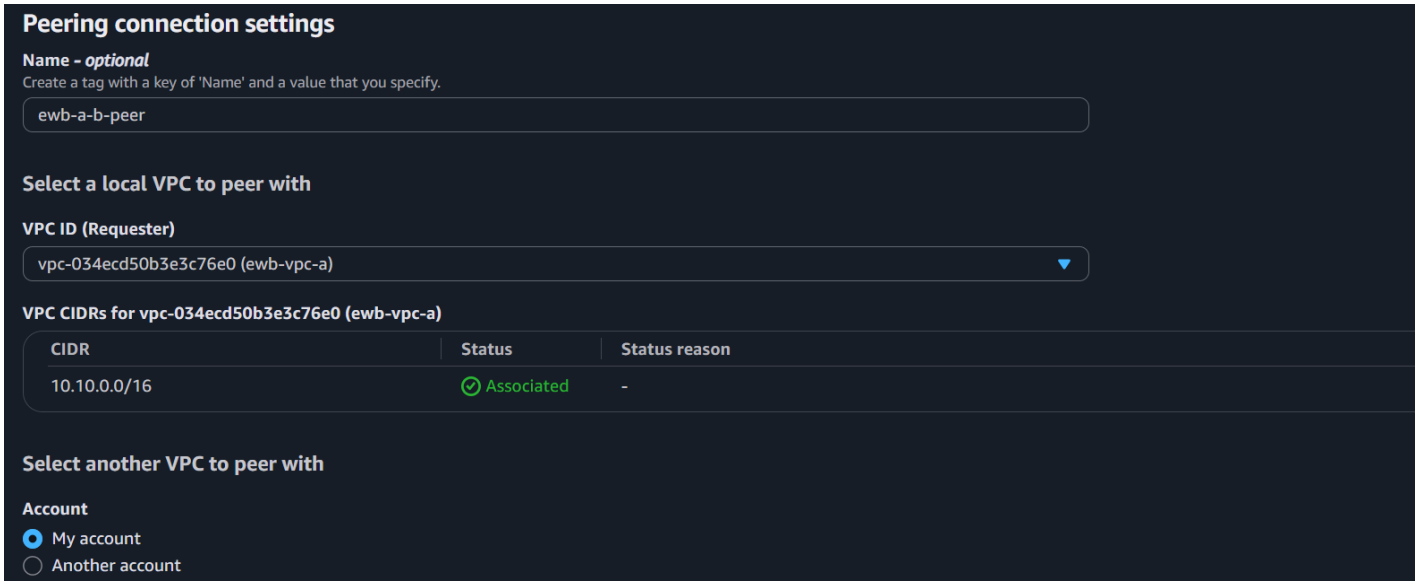
```
[ec2-user@ip-10-10-2-26 ~]$ ping -c 4 10.20.1.28
PING 10.20.1.28 (10.20.1.28) 56(84) bytes of data.

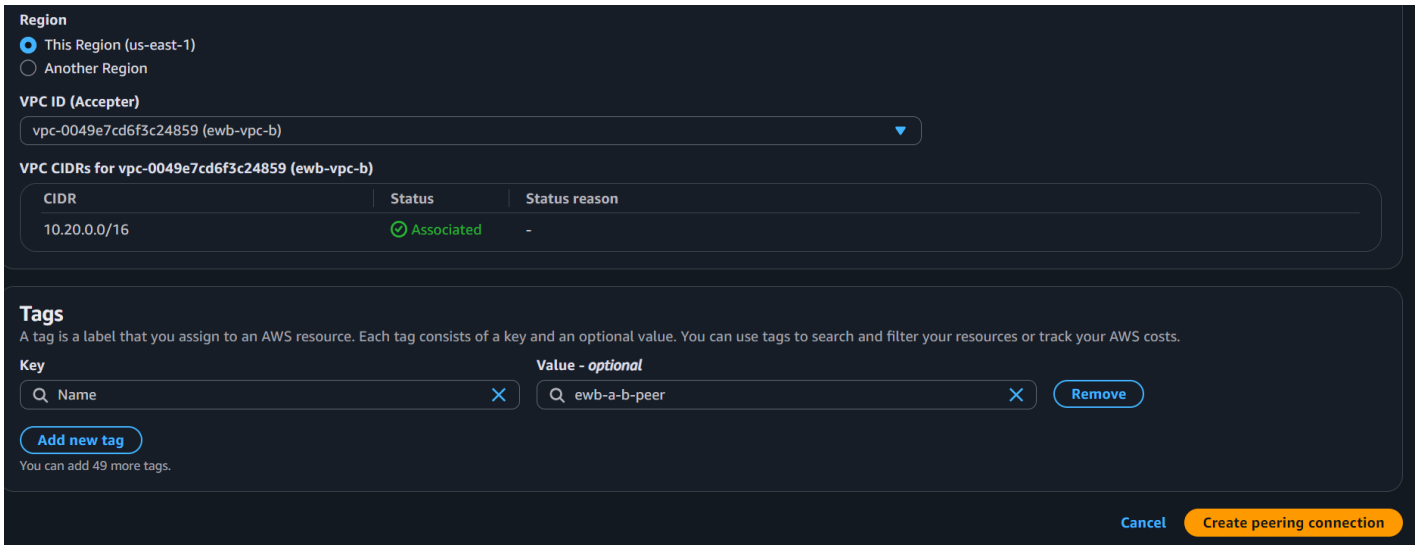
--- 10.20.1.28 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3103ms
```

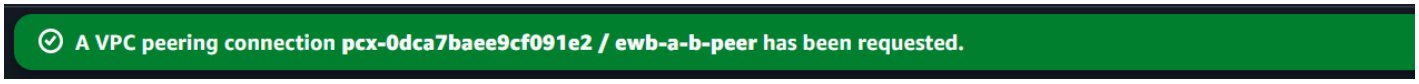
Step 33:- Create the VPC Peering Connection.



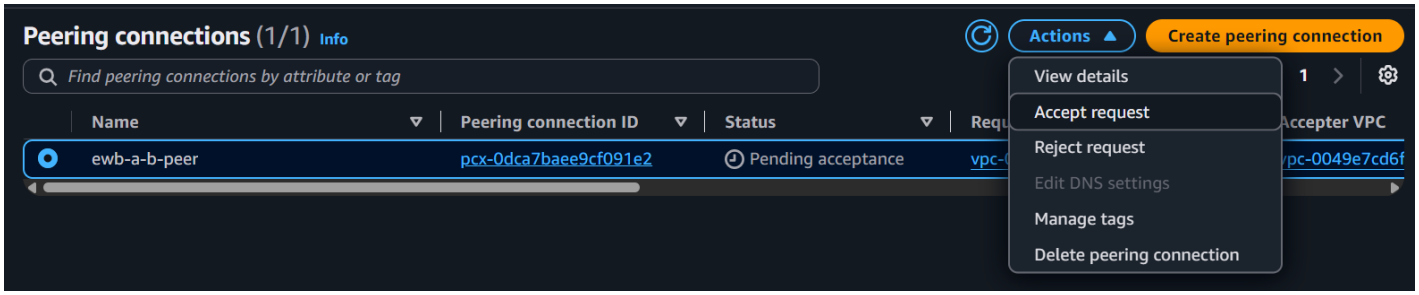


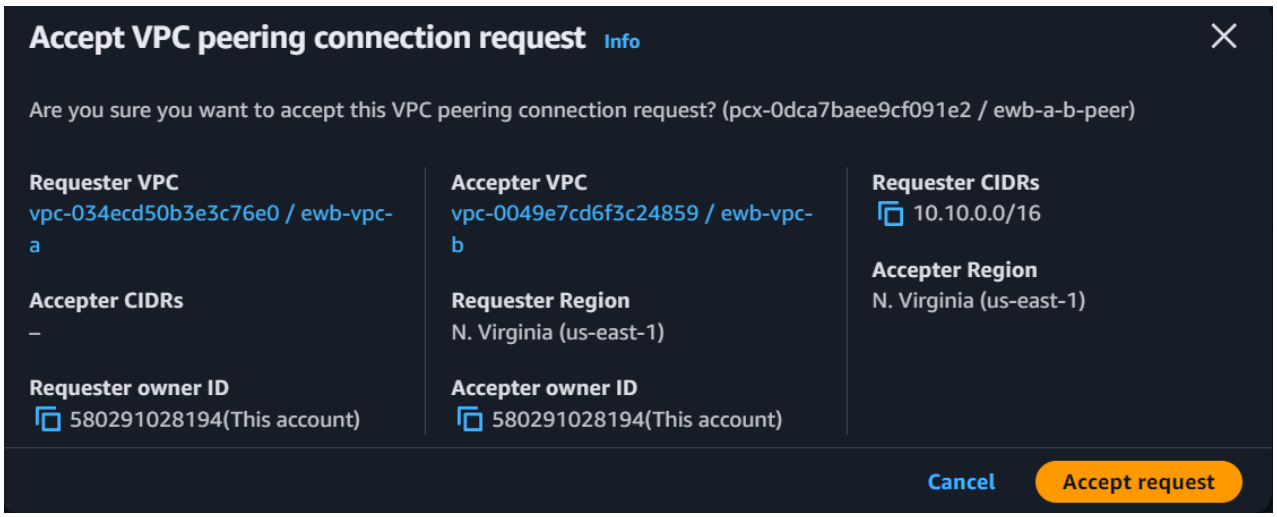


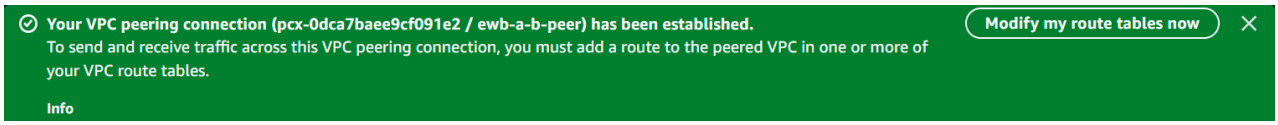




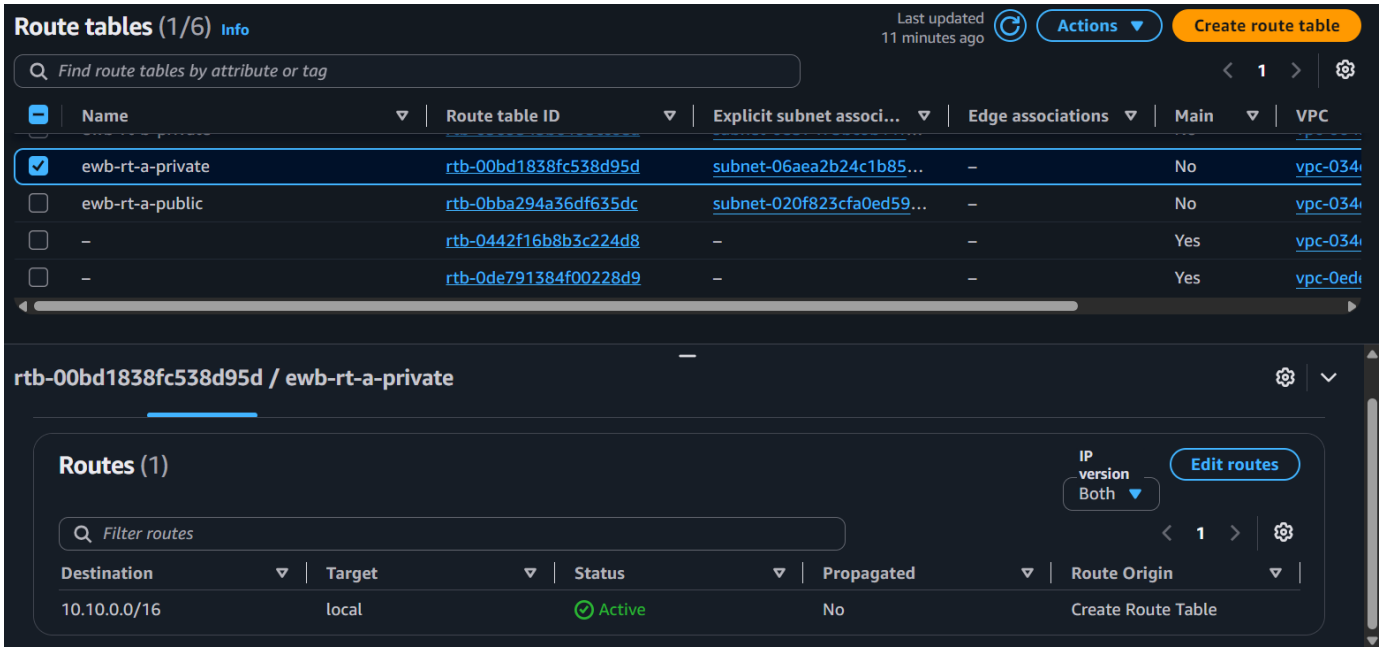
Step 34:- Accept the VPC Peering Connection.

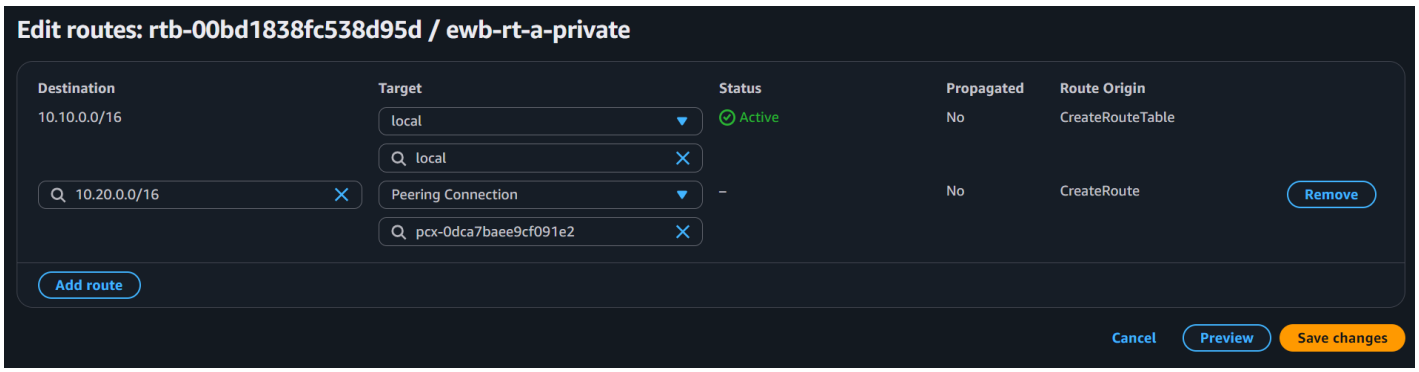


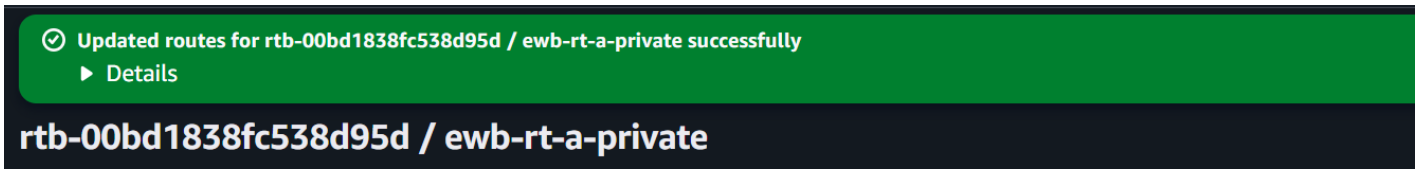




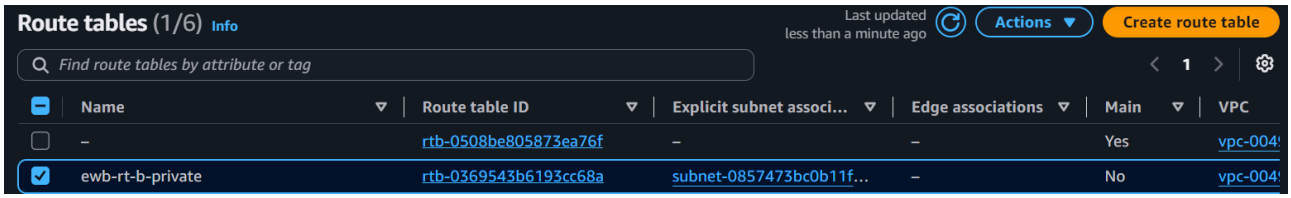
Step 35:- Add the first route.

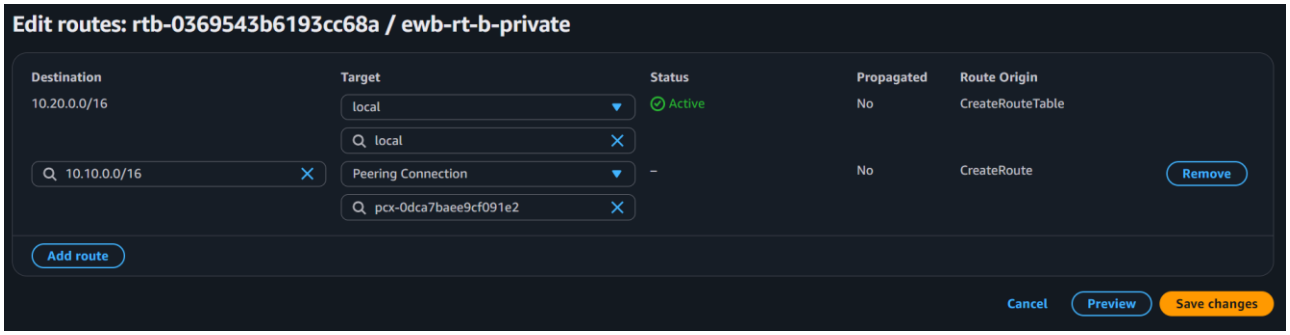


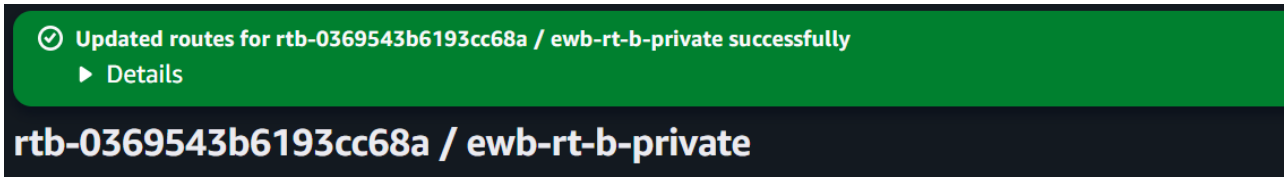




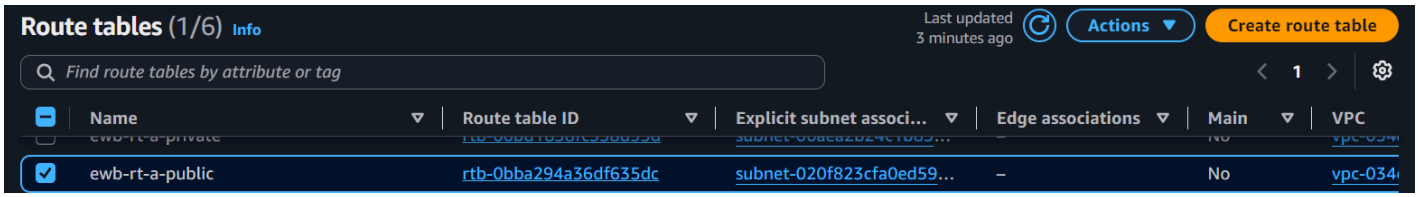
Step 36:- Add the second route.







Step 37:- Add the public route.





Step 38:- Test connectivity from App B.

```
[ec2-user@ip-10-10-2-26 ~]$ ping -c 4 10.20.1.28
PING 10.20.1.28 (10.20.1.28) 56(84) bytes of data.
64 bytes from 10.20.1.28: icmp_seq=1 ttl=127 time=0.586 ms
64 bytes from 10.20.1.28: icmp_seq=2 ttl=127 time=0.174 ms
64 bytes from 10.20.1.28: icmp_seq=3 ttl=127 time=0.154 ms
64 bytes from 10.20.1.28: icmp_seq=4 ttl=127 time=0.161 ms

--- 10.20.1.28 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3159ms
rtt min/avg/max/mdev = 0.154/0.268/0.586/0.183 ms
[ec2-user@ip-10-10-2-26 ~]$
```

Step 39:- Test connectivity in the reverse direction.

```
[ec2-user@ip-10-10-2-26 ~]$ exit
logout
Connection to 10.10.2.26 closed.
[ec2-user@ip-10-10-1-105 ~]$
```

```
[ec2-user@ip-10-10-1-105 ~]$ ssh ec2-user@10.20.1.28
The authenticity of host '10.20.1.28 (10.20.1.28)' can't be established.
ED25519 key fingerprint is SHA256:DicNL67WqYQLtgtkJUE/xeu4aQAeZyx3LJfdsimDOD0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.28' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-28 ~]$
```

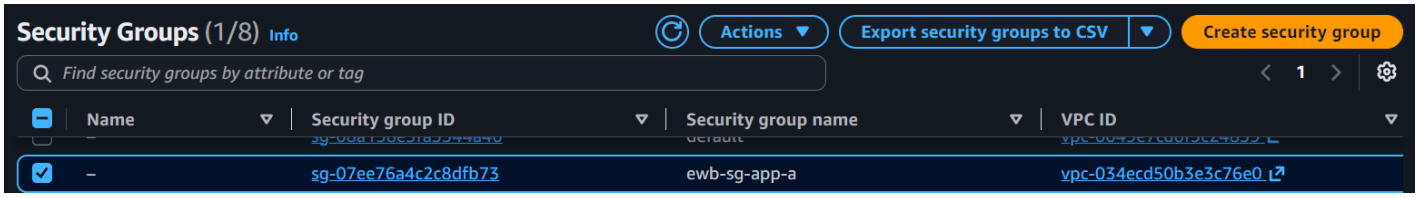
```
[ec2-user@ip-10-20-1-28 ~]$ ping -c 4 10.10.2.26
PING 10.10.2.26 (10.10.2.26) 56(84) bytes of data.
64 bytes from 10.10.2.26: icmp_seq=1 ttl=127 time=0.159 ms
64 bytes from 10.10.2.26: icmp_seq=2 ttl=127 time=0.210 ms
64 bytes from 10.10.2.26: icmp_seq=3 ttl=127 time=0.191 ms
64 bytes from 10.10.2.26: icmp_seq=4 ttl=127 time=0.171 ms

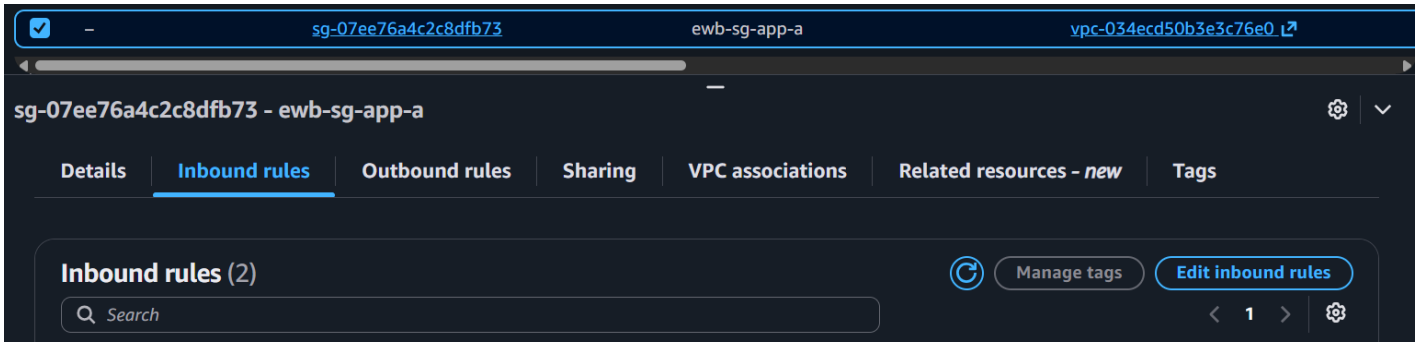
--- 10.10.2.26 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3121ms
rtt min/avg/max/mdev = 0.159/0.182/0.210/0.019 ms
[ec2-user@ip-10-20-1-28 ~]$
```

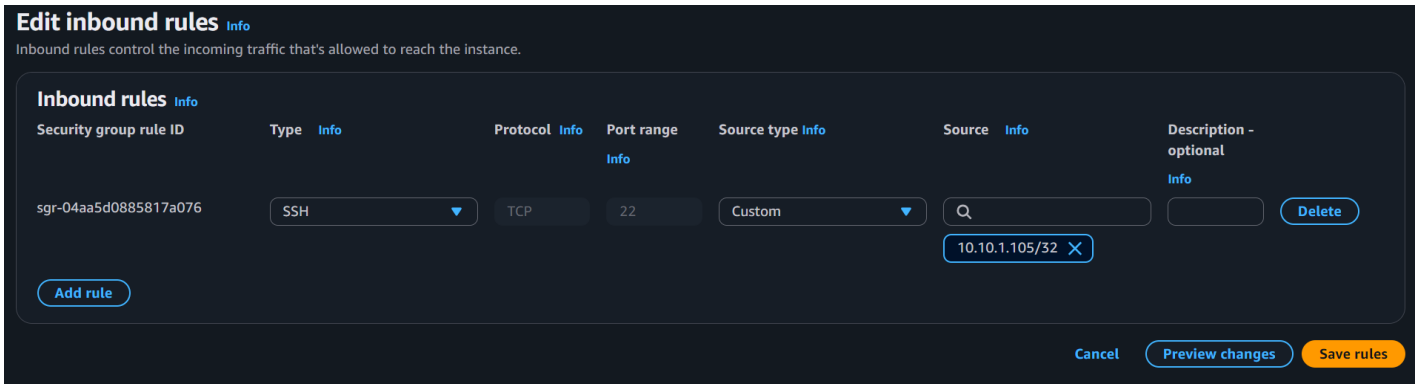
Step 40:- Start the controlled ICMP test.

```
[ec2-user@ip-10-20-1-28 ~]$ exit
logout
Connection to 10.20.1.28 closed.
```

Step 41:- Temporarily block ICMP traffic.







Step 42:- Test the expected ping failure.

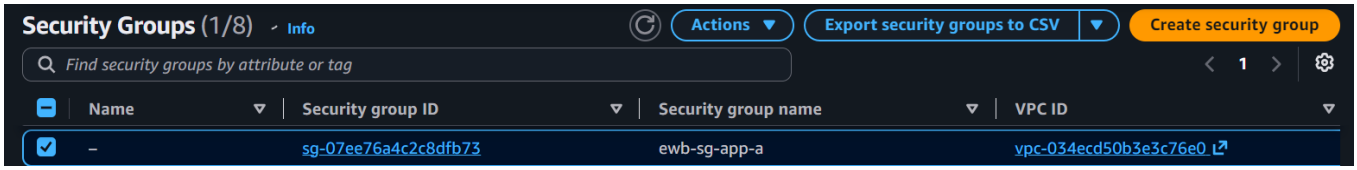
```
[ec2-user@ip-10-10-1-105 ~]$ ssh ec2-user@10.20.1.28
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 13:01:52 2026 from 10.10.1.105
[ec2-user@ip-10-20-1-28 ~]$
```

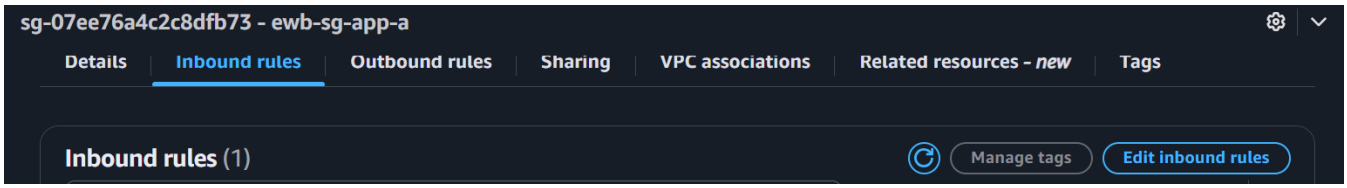
```
[ec2-user@ip-10-20-1-28 ~]$ ping -c 4 10.10.2.26
PING 10.10.2.26 (10.10.2.26) 56(84) bytes of data.

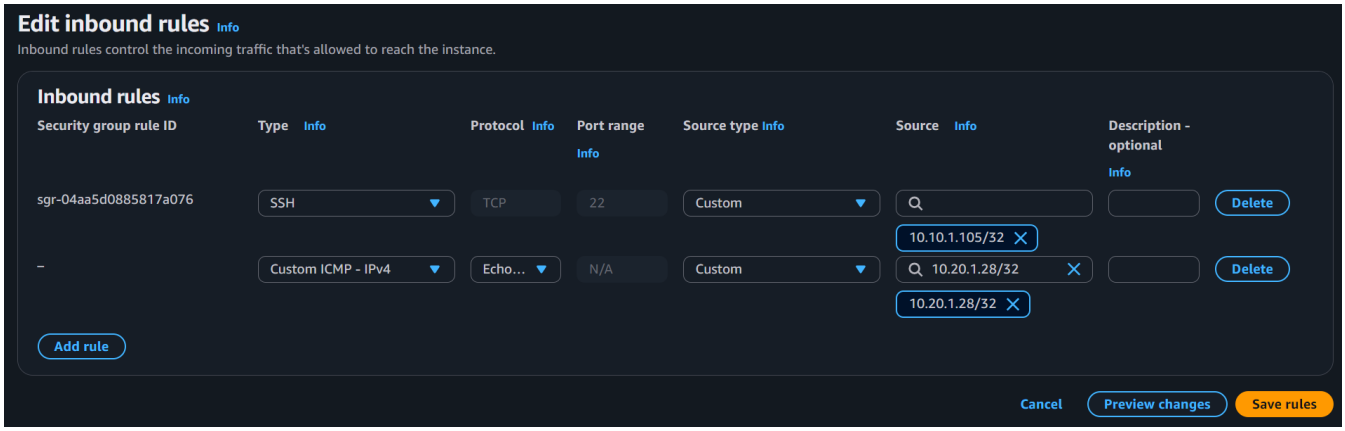
--- 10.10.2.26 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3084ms

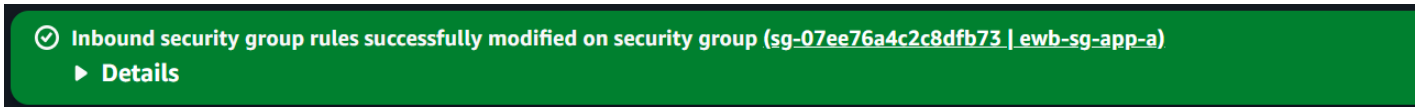
[ec2-user@ip-10-20-1-28 ~]$
```

Step 42:- Restore the ICMP rule.









Step 43:- Verify that connectivity is restored.

```
[ec2-user@ip-10-20-1-28 ~]$ ping -c 4 10.10.2.26
PING 10.10.2.26 (10.10.2.26) 56(84) bytes of data.
64 bytes from 10.10.2.26: icmp_seq=1 ttl=127 time=0.170 ms
64 bytes from 10.10.2.26: icmp_seq=2 ttl=127 time=0.181 ms
64 bytes from 10.10.2.26: icmp_seq=3 ttl=127 time=0.192 ms
64 bytes from 10.10.2.26: icmp_seq=4 ttl=127 time=0.199 ms

--- 10.10.2.26 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3152ms
rtt min/avg/max/mdev = 0.170/0.185/0.199/0.011 ms
[ec2-user@ip-10-20-1-28 ~]$
```